OSM+: Billion-Level OpenStreetMap Dataset for City-wide Experiments

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Abstract

Road network data provides rich information about cities, but processing worldwide OpenStreetMap (OSM) data is computationally intensive, and the resulting graphs are often difficult to unify for benchmarking downstream tasks. Existing graph learning benchmarks fail to capture the billion-scale and unique topological properties of real-world road networks, leaving model scalability underexplored. To close this gap, we process OSM data with distributed cloud computing using 5,000 cores and release **OSM+**, a structured worldwide 1-billion-vertex road network graph dataset designed for high accessibility and usability. OSM+ is open source and globally downloadable, providing an open-box graph structure together with an easy spatial query interface that allows users to retrieve, inspect, and integrate road network topology, geometry, and attributes without repeatedly preprocessing raw OSM files. We demonstrate the utility of OSM+ through four illustrative use cases: basic query, city boundary detection, traffic prediction, and traffic policy control. For traffic prediction, we construct a new 31-city benchmark by processing traffic data and combining it with OSM+, enabling broader spatial coverage and more comprehensive evaluation than commonly used datasets, while scaling from hundreds of road network intersections to thousands. For traffic policy control, we release a new six-city dataset at a much larger scale, introducing challenges for thousand-scale multi-agent coordination and controller scalability. We also provide data processing tools for integrating multimodal spatial-temporal data with OSM+ for geospatial foundation model training, thereby expediting the discovery of compelling scientific insights.

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1. Introduction

Road networks form the physical skeleton of cities, connecting neighborhoods, functional areas, and even different cities into an integrated spatial system. Over long periods of urban development, road networks and urban regions have co-evolved: the expansion of roads shapes land development and accessibility, while emerging residential, commercial, and industrial areas in turn drive the construction and reconfiguration of roads. As a result, the structure of a road network provides rich signals about the spatial layout, landscape patterns, and functional organization of a city. It also plays a fundamental role in shaping human mobility, since travel demand, route choice, and accessibility are all constrained by the underlying road infrastructure. For instance, prior studies have reported a strong correlation between human mobility intensity and road network density (Zhu et al., 2022). Therefore, investigating road network structures is a fundamental prerequisite for a wide range of urban studies, including urban planning, transportation analysis, traffic prediction, accessibility evaluation, and urban representation learning.

However, obtaining accurate and fine-grained road network data for open public research remains challenging. High-quality road network datasets are often collected, maintained, and updated at substantial cost by commercial map providers such as Google (Google Maps), Bing (Bing Maps), Baidu (Baidu Maps), and Gaode (Gaode Maps). Consequently, these platforms are primarily designed to support commercial applications and typically expose only limited high-level APIs to the public, such as point-of-interest search, geocoding, and origin-destination route planning. While these services are useful for application-level development, they are insufficient for many research scenarios that require direct access to the underlying road topology, geometry, and attributes. The restricted accessibility of commercial map data therefore creates a substantial barrier to open, reproducible, and large-scale urban research.

Open-source map platforms such as OpenStreetMap (Haklay & Weber, 2008a), built through crowdsourcing by contributors worldwide, provide a promising alternative to commercial map services. OpenStreetMap offers broad spatial coverage and rich map attributes, making it an important

public source for road network research. Nevertheless, directly using OpenStreetMap data in scientific experiments remains challenging. The raw data are massive in scale and stored in a complex geospatial format involving nodes, ways, relations, and heterogeneous tags. Converting these raw map objects into a clean, connected, and analysis-ready road network graph requires non-trivial preprocessing, including filtering road-related objects, repairing invalid geometries, connecting broken edges, removing redundant segments, and transforming the data into task-specific formats. This process often involves multiple rounds of trial and error, making it both time-consuming and labor-intensive.

Figure 1 illustrates this gap by comparing several common ways of querying OpenStreetMap data. Website-based search is convenient for manual inspection but lacks explicit graph structures for computation. The Overpass API provides flexible access to map objects, but users still need to write queries and process the retrieved raw data. OSMnx simplifies graph extraction through a Python interface, yet large-scale queries can be slow because graphs are constructed on demand. In contrast, OSM+ provides a precomputed graph-structured dataset stored in a cloud database, enabling efficient and scalable queries over road network topology, geometry, and attributes. Therefore, it is desirable to provide *an intermediate, processed, and reusable format of road network data* that can support diverse downstream applications and accelerate scientific discovery.

Following this direction, several studies (Grinberger et al., 2022; Bartzokas-Tsiompras, 2022; Ding et al., 2022) have released tools for cleaning and processing OpenStreetMap road network data. However, important issues remain. OpenStreetMap preprocessing still involves a complex pipeline of format conversion, geometric transformation, topology construction, and spatial aggregation. Even processing a region of about 1,000 km^2 may take around 10 hours on a machine with 32 CPU cores and 128GB memory. Scaling to the global level is even more challenging: the worldwide raw OpenStreetMap data are roughly 1.1TB, and the processed graph can contain more than one billion vertices. Although region-wise processing can reduce memory pressure, it requires additional distributed designs and may introduce boundary inconsistencies when merging results. Consequently, researchers often build their own pipelines, producing incompatible road network datasets from the same raw source. This hinders reproducible comparison and large-scale integration, especially for multi-modal urban foundation models. Therefore, it is important to *build a pre-processed, structured, and publicly available version of OpenStreetMap* as a unified infrastructure for open urban research.

In this paper, we introduce OSM+, a dataset, benchmark, and system resource that makes OpenStreetMap usable for

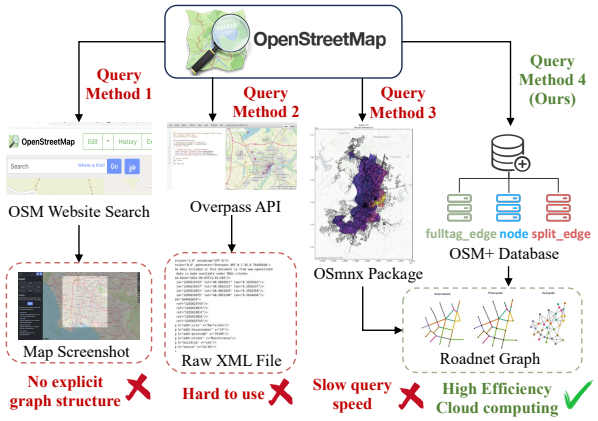


Figure 1. Comparison of common OpenStreetMap query paradigms and the proposed OSM+ dataset.

large-scale city-wide experiments. OSM+ consists of four released components. **First**, we release six core OSM+ data tables. Among these six tables, three contain comprehensive non-road information, such as POI nodes, building footprints, and administrative boundaries, while the other three contain the road-network graph used in most experiments. **Second,** we release an automated preprocessing pipeline that transforms raw Geofabrik/OpenStreetMap mirrors into structured files, making the construction process reproducible. **Third**, we provide efficient query tools based on the *CityBrain Open Platform* for managing OSM+ and extracting spatial subgraphs at scale. **Fourth**, we release benchmark code, processed benchmark graphs, experimental configurations, and converters for downstream tasks.

The contributions of this paper can be summarized as follows:

- We introduce OSM+, an open-source, worldwide billion-scale road-network graph dataset with an open-box graph structure and an easy spatial query interface. To our knowledge, OSM+ is the first billion-scale graph resource that enables studying the unique topological properties of real-world road networks and exposes critical scalability limitations of modern GNNs. Detailed instructions for accessing the OSM+ dataset are shown in Appendix A.
- We provide a cloud-computing-based platform to enable efficient billion-scale graph query and processing, as well as an automated pipeline for transforming raw OpenStreetMap files into structured vertex and edge tables. The efficient query interfaces are available at **CityBrain Open Platform**, while the related code, preprocessing scripts, benchmark code and tutorials are available at our official GitHub repository: `https://github.com/SJTU-CILAB/OSM-dataset`.
- We provide four example application scenarios based on the

OSM+ dataset: basic query, city-boundary detection, traffic prediction, and traffic policy control. The released data converters show how OSM+ can serve as a road-network backbone for integrating multi-modal spatial-temporal data and for constructing reproducible downstream tasks.

- For the traffic-prediction task, we construct a new 31-city offline research benchmark by aligning public historical traffic-flow observations from UTD19 with OSM+ road networks. For traffic policy control, we release six city-scale simulation scenarios with up to 18,948 intersections. These resources raise new challenges for both predictive accuracy and scalability, while source-specific limitations of the third-party upstream datasets are explicitly documented.

2. OSM+: Structured Road Network Dataset with Computing Support

To address the challenges of raw OSM data usability, we present OSM+, a scalable system that transforms raw map data into structured, analysis-ready road networks. Figure 2 illustrates the overall construction and query pipeline of OSM+. In the offline stage, raw OpenStreetMap data from Geofabrik and the OpenStreetMap API are first filtered to extract road-related linestring objects as initial road candidates. These objects are then cleaned through distributed processing to remove invalid, redundant, and inconsistent road segments. After cleaning, we split road geometries at intersections and endpoints to define roadnet vertices and edges, and store the processed results as structured OSM+ tables, including both full OSM tables and roadnet-only graph tables. Once this one-time preprocessing is completed, users no longer need to repeatedly download and process raw OSM files on local machines. Instead, they can directly issue SQL-style spatial queries to OSM+ and obtain ready-to-use road network graphs efficiently.

2.1. OSM+ Data Model and Roadnet Graph

Many downstream machine learning tasks in transportation rely on graph-structured road networks. OSM+, however, is not released as a single road graph table, but as six structured public tables deployed on *MaxCompute*, formerly known as *ODPS*, a distributed data warehousing platform for SQL-style analysis over terabyte-to-petabyte-scale data. Among these tables, three full OSM tables preserve cleaned map objects and attributes, such as point records and tagged way-like objects, while three roadnet-only tables provide the compact road-network graph used in most downstream experiments. The roadnet-only tables are obtained by filtering road-related objects and splitting road geometries into elementary start–end segments. This organization allows users to either access rich map context, such as POIs, buildings, and administrative boundaries, or directly query a cleaned roadnet graph for transportation tasks. We define the roadnet graph as follows.

Definition 2.1 (OSM+ Roadnet Graph)**.** Given the roadnet-only tables in OSM+, we define the road network as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

(1) ***Roadnet vertex****: Each vertex represents a topological point on the road network, such as a road intersection, a road endpoint, or a split point generated when a road geometry is divided into elementary segments. Each vertex is associated with spatial coordinates and a unique identifier.*

(2) ***Roadnet edge****: Each edge represents an elementary road segment connecting two roadnet vertices. It is derived from a split OSM road geometry and preserves references to the original OSM object and its attributes, such as road category, geometry, and tags when available.*

Under this definition, POIs, buildings, and administrative boundaries are not treated as roadnet vertices. Instead, they are auxiliary map entities stored in the full OSM+ tables and can be joined with the roadnet graph through spatial predicates or object identifiers. For example, a SQL query can retrieve all road segments within a specified bounding box and join them with nearby POI records to compute road-level features.

2.2. Information of OSM+

Statistics. In the original OpenStreetMap data, each edge is represented as a linestring. We subdivide each linestring into several split edges, each with exactly one start point and one end point. The split edges that constitute the road network are referred to as "split edges on roadnet". As shown in Table 1, the worldwide road network forms a massive graph with billions of vertices and edges. Moreover, the spatial density of vertices can serve as a proxy for population distribution and urbanization levels.

Table 1. Basic statistics of OSM+ database.

Subset	Counts		
	Vertices	Split Edges	Fulltag Edges
Roads and POI	9,643,837,963	10,656,542,261	1,110,784,986
Roadnet Only	1,901,611,316	2,021,159,972	202,281,206

Spatial distribution & categories. The spatial distribution of road data is imbalanced. Figure 3 reports the numbers of road-network vertices and edges across continents. Europe has the largest number of vertices and edges, followed by Asia, while Oceania has the smallest. The distribution of road categories is shown in Figure 4. It can be observed that the number of roads increases approximately exponentially as the road hierarchy level decreases, providing a compact view of the hierarchy-tag distribution inherited from OSM.

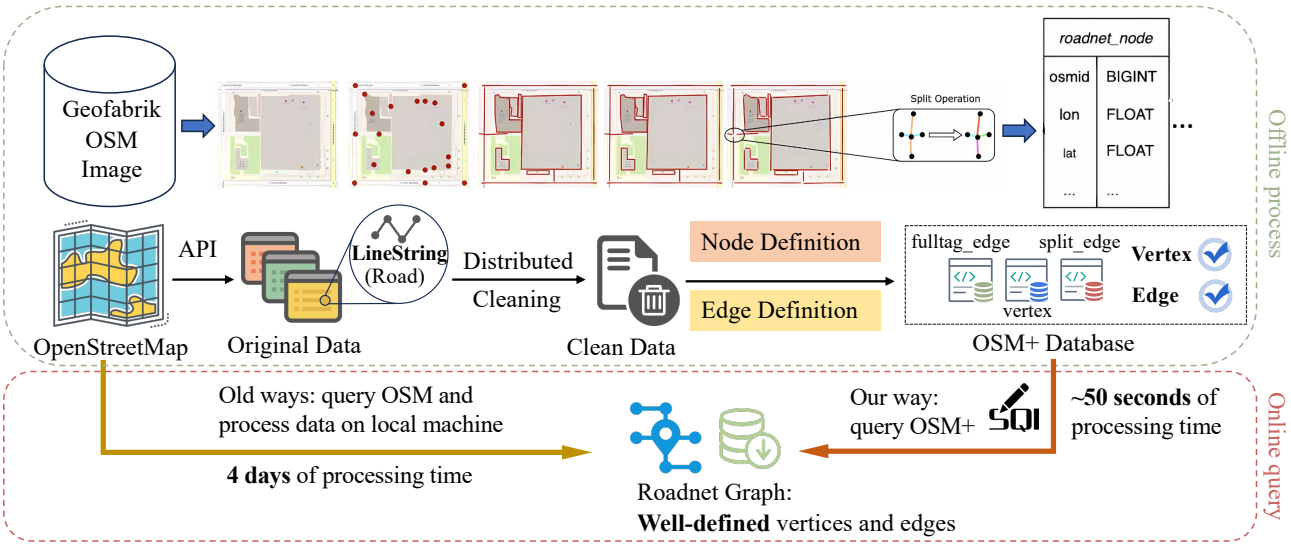


Figure 2. OSM+ database provides worldwide edges and points. Only basic SQL is needed to facilitate the efficient construction of urban graph structures at a global scale.

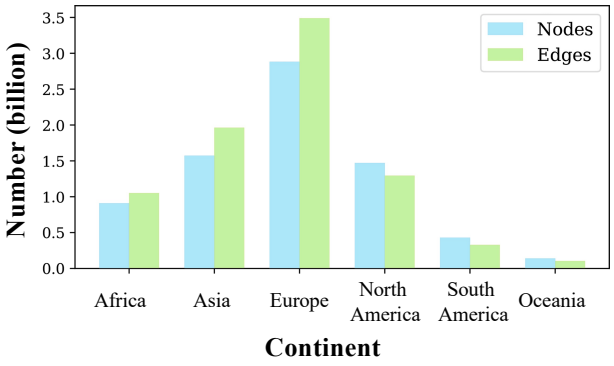


Figure 3. Basic continent-level statistics of the OSM+ database, showing the number of road-network nodes and edges across different continents.

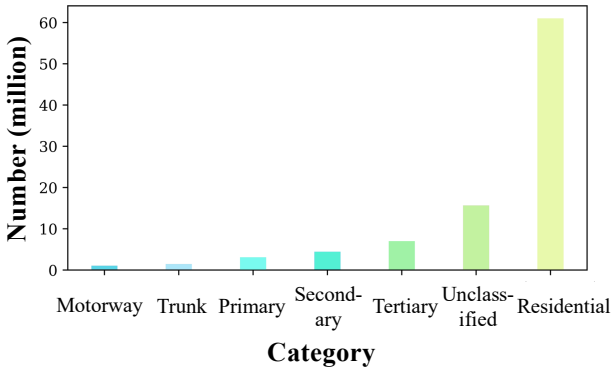


Figure 4. Category-level statistics of the OSM+ database, illustrating the distribution of road segments across major road types.

3. Basic Usage of OSM+

3.1. Efficient Spatial Computing Support

Given that OSM+ encodes a global road network with billions of vertices and edges, executing interactive queries on a single commodity server is impractical. We therefore deploy OSM+ on cloud-computing resources and design parallel query mechanisms to enable efficient large-scale access to the data. Figure 5 illustrates the difference between the conventional local processing workflow and the cloud query workflow based on OSM+. In the local workflow, users typically need to download and process large raw OSM tables, perform expensive join operations, and manually handle sharding, distribution, and data-precision issues. When the target region is large or the join involves global-scale tables, this process can easily exceed the memory capacity of a local machine and lead to out-of-memory errors. Moreover, even if only a specific regional map is needed, the local workflow still requires cumbersome preprocessing over a much larger data scope before the required subset can be extracted.

In contrast, OSM+ organizes the processed global road-network data in the cloud and exposes efficient query interfaces. Users only need to specify the query condition, such as longitude-latitude coordinates or a target spatial region, and the system automatically locates the relevant shards, executes the query in parallel, and returns the required data. This design avoids repeated local preprocessing, reduces the memory burden on users, and makes large-scale spatial access to OSM+ substantially easier and more efficient. Here, we use several basic query operations as examples to demonstrate how OSM+ supports practical access to the

global road-network database.

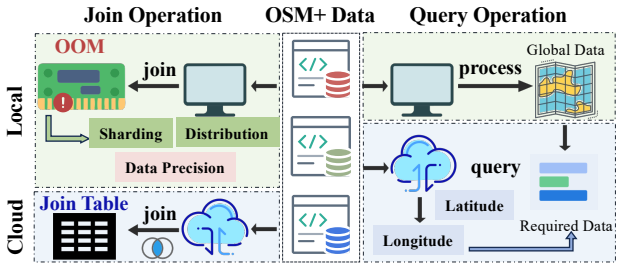


Figure 5. Comparison between conventional local OSM processing and the OSM+-based cloud query workflow.

Point Query Operation. Point query is one of the fundamental spatial operations supported by OSM+. Given a query point and a radius, the goal is to select all vertices in the `osm_node` table that fall within the specified radius. We apply window-check optimization that partitions the geographic space into a uniform grid at a specified spatial resolution. During query processing, we restrict the search to vertices located in the grid cell containing the query point and its neighboring cells, thereby pruning a large fraction of irrelevant vertices. As shown in Table 2, we conduct 1,000-point queries with and without the window-check optimization on a commodity cloud-computing cluster, and the results indicate that window check substantially improves point-query efficiency.

Nearest-neighbor Query Operation. Given a query point and a latitude-longitude bounding box, the goal is to find the nearest vertex (nearest-neighbor query) or the k nearest vertices (k-nearest-neighbor query) within this region. To reduce query time, we employ k-d tree, a binary space-partitioning tree that recursively divides a k-dimensional space. As shown in Table 2, the k-d tree implementation substantially reduces the running time required for these queries compared with a linear scan baseline.

Table 2. Query efficiency comparison between with or without "window check" and "k-d tree" optimization technique.

Method	Point Query		Nearest Pair Query	
	Runtime(s)	Core × min	Runtime(s)	Core × min
W/O Optimization	310	2.06	41.14	0.05
With Optimization	**79**	**1.33**	**1.49**	**0.01**

Table 3. Efficiency of global KDE estimation on different computing clusters.

Platform	Sample Rate					
	1/1,000		1/10,000		1/100,000	
	Runtime(s)	Core × min	Runtime(s)	Core × min	Runtime(s)	Core × min
ECS	OOM	OOM	841.62	14.02	33.06	0.22
Spark	29,014.66	17,408.49	197.02	118.21	7.21	4.32
ODPS	**72.13**	**1.02**	**34.26**	**0.30**	**6.04**	**0.10**

Using these techniques, we build an example comprehensive calculation task to conduct the KDE kernel density estimate of each intersection vertex on the global road network data, illustrating why it is necessary to run experiments on cloud computing. We use three different sampling rates to sample the original global road network data and compare the runtime and memory cost on different platforms. The experimental results are shown in Table 3. It is observed that by utilizing the ODPS computing engine, we could employ an optimized query algorithm which significantly outperforms that of other computing clusters in terms of both runtime and memory utilization.

3.2. Use Case: City Boundary Detection

Accurately delineating city boundaries is important for urban planning, public resource allocation, and location-based services. However, administrative boundaries do not always reflect the functional extent of cities, especially in rapidly urbanizing regions where economic activities and commuting flows may extend beyond official borders. Since urban areas are typically characterized by dense and interconnected road networks, road-network density provides a useful signal for identifying functional city boundaries.

In this use case, we show how OSM+ can be used to derive functional city boundaries from global-scale road-network data. Specifically, we cluster regions according to road-network structure and density, and compare the resulting clustering-based boundaries with official administrative boundaries. By overlaying the two types of boundaries, both consistent regions and mismatched areas can be identified. Large discrepancies indicate that the effective urban footprint has expanded beyond administrative limits or that one administrative unit contains multiple relatively separated urbanized areas.

Figure 6 presents representative results in central Europe, the east coast of the United States, and the Yangtze River Delta in China. Colored polygons denote the clustering-based city regions, while boundary lines indicate administrative divisions. In many cases, such as Shanghai and Jiaxing in China, the clustering results are broadly consistent with administrative boundaries. In other cases, an administrative region may contain multiple clusters, suggesting several disconnected urban centers within the same jurisdiction. Mismatches are often observed near administrative borders where dense road networks continue across official boundaries, reflecting cross-border urban expansion or strong regional connectivity.

To further validate the road-density-based interpretation, we compare road networks with population and nightlight data in Figure 7. The results show that areas with denser roads generally coincide with higher population density and stronger night-time light intensity, suggesting that road-network density is a reasonable proxy for urban activity.

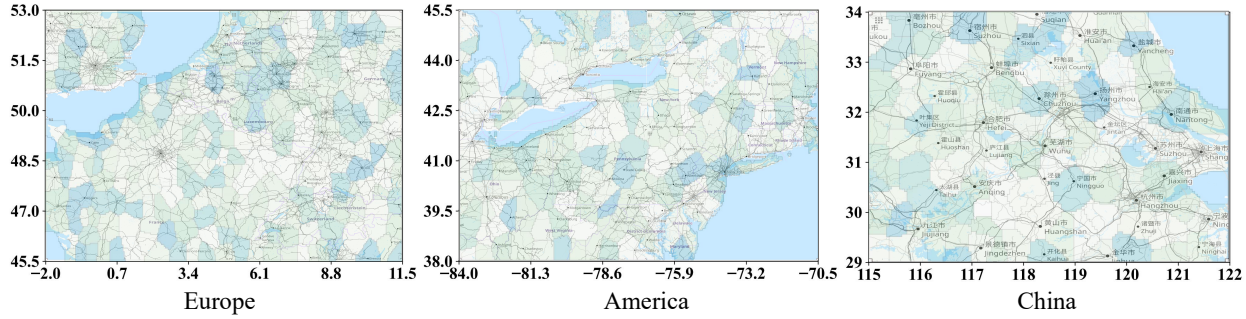


Figure 6. The city boundary map obtained from the clustering results of central Europe, the east coast of the United States and the Yangtze River Delta in China is overlaid with the road maps of the three regions on OpenStreetMap, and latitude and longitude are added to distinguish them.

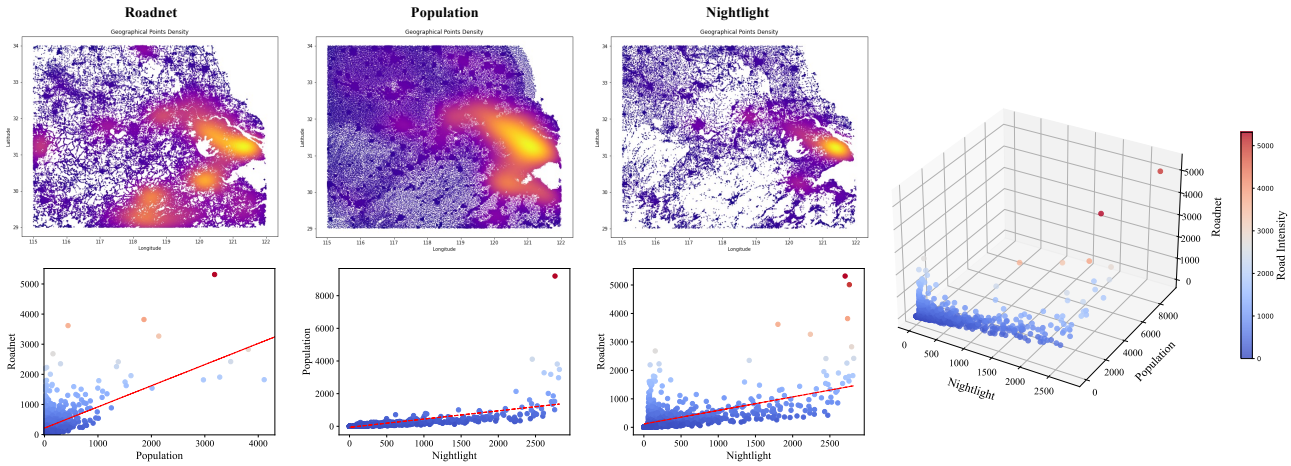


Figure 7. Comparing the road, population and nightlight data map in the same area, the road density is positively correlated with the population and night lighting data density.

In addition, population maps can be thresholded by high red-channel values in RGB space to approximate city-center regions, while nightlight maps can be filtered by brightness to extract highly active urban cores. These comparisons indicate that OSM+ can support scalable and interpretable city boundary detection by providing a unified road-network foundation.

4. OSM+ Assisted Tasks

In this section, we introduce two typical applications of OSM+ combined with other datasets, which are **traffic prediction** and **traffic policy control**. As summarized in Figure 8, OSM+ provides a unified data foundation for downstream urban and transportation applications. Starting from the OSM+ database, the pipeline first uses task-specific data converters to transform raw road-network records and auxiliary urban information into application-ready inputs. The prepared data can then be directly consumed by downstream systems for traffic simulation, supporting tasks including traffic signal control and congestion pricing. In this way, OSM+ bridges large-scale road-network data and practical downstream tasks through a standardized conversion-and-application workflow.

4.1. City-scale Traffic Prediction

Traffic prediction has been extensively studied due to its central role in intelligent transportation systems. Existing work commonly evaluates models on standard benchmarks such as METR-LA and the PEMS series (Li et al., 2017; Jiang & Luo, 2022). However, recent studies have observed that, on these fixed benchmarks, many newly proposed methods yield only marginal gains over strong baselines, raising concerns about benchmark saturation and potential overfitting to a limited set of evaluation scenarios (Oakley et al., 2024).

To address this limitation, we introduce a new benchmark consisting of **31 city-level traffic datasets**, each paired with traffic flow observations and the corresponding road-network structure. Compared with existing benchmarks, our OSM+-based benchmark cover a substantially broader range of cities and introduce several distinctive challenges,

Table 4. Basic statistics of traffic signal control experiments.

Indicator	Former datasets		OSM+ dataset					
	Hangzhou	Manhattan	New York	Los Angeles	Beijing	Shanghai	London	Paris
Intersections	16	2,510	5,971	6,663	18,948	14,750	5,895	1,721
Vehicles	2,983	48,079	90,059	112,291	130,851	85,480	107,105	101,929

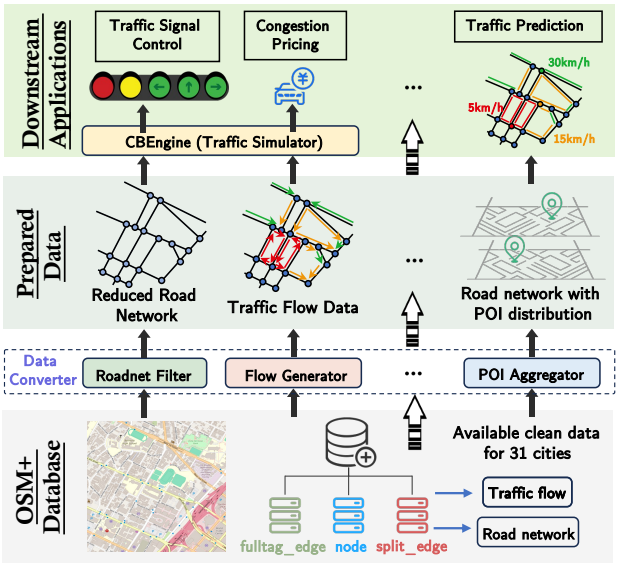


Figure 8. The framework of the downstream task supporting OSM+ database.

as summarized in Figure 9.

- **Heterogeneous urban traffic dynamics.** Unlike conventional benchmarks that are often dominated by homogeneous highway or arterial-road settings, our datasets cover diverse in-city traffic scenarios with different road functions, spatial layouts, intersection densities, and local traffic patterns. This heterogeneity better reflects real urban environments and provides a more realistic test bed for evaluating model generalization.

- **Sparse sensing over dense road networks.** In our datasets, observed traffic sensors are sparse relative to the dense city-scale road networks. Models must therefore infer traffic dynamics from limited and unevenly distributed measurements, which is common in real deployments due to cost, maintenance, and infrastructure constraints. This setting encourages models that are robust to partial observations and can exploit broader road-network structure.

- **City-scale spatial coverage.** Our OSM+ benchmark operates at a much larger spatial scale than widely used traffic prediction datasets. While METR-LA and PEMS typically contain only a few hundred sensor nodes, our 31 city-level datasets cover large urban road networks with far more intersections and segments, with the largest graph containing approximately 30,000 nodes. This enables evaluation of both predictive accuracy and scalability for city-wide deployment.

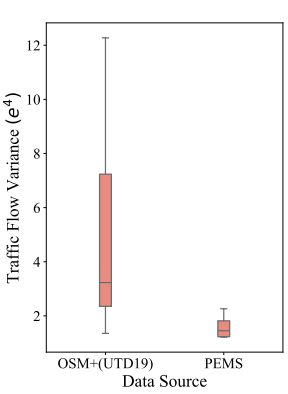
To evaluate the performance, we test seven frequently-cited baseline methods on these 31 cities. All models are evaluated on three different prediction horizons (3, 6, and 12) following the widely used setting. The average results over prediction horizons are shown in Table 5, and the full per-horizon results with standard deviations are shown in Appendix F. Details of all baseline methods are shown in Appendix E.3.

The results confirm that no single baseline dominates all cities and metrics. Crossformer, GWNet, MTGNN, and DLinear each obtain the best results on different cit–metric pairs, while performance varies more strongly on OSM+(UTD19) than on conventional saturated datasets. Extreme MAPE values are often associated with low-flow periods, heterogeneous city-level distributions, or sensor-to-road alignment noise. These phenomena are part of the intended stress-test value of OSM+: they reveal scalability and robustness bottlenecks that are less visible on small traffic benchmarks.

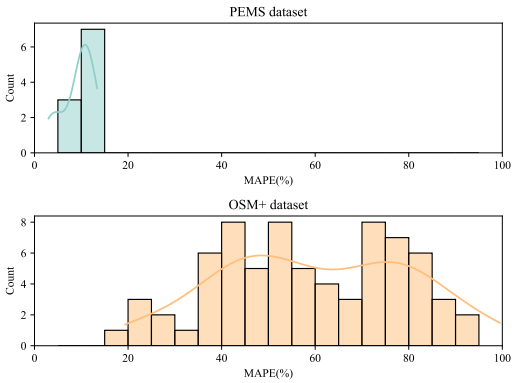
4.2. Traffic Policy Control

Starting from the cleaned OSM+ road network, we construct microscopic simulation environments for traffic policy control experiments, such as traffic signal control. As summarized in Figure 8, the pipeline converts OSM+ road graphs into simulator-ready road networks, generates traffic demand, and configures the corresponding simulation files. This process enables OSM+ to serve not only as a static road-network database, but also as a source for building executable traffic-control benchmarks.

Compared with previously used datasets for traffic signal control experiments (Wei et al., 2019), our pipeline supports scenarios with more extensive spatial coverage and more realistic city-scale network structures. In this work, we release six representative city scenarios that have been carefully cleaned and validated. Table 4 reports their basic statistics, including the number of signalized intersections and simulated vehicles. Figure 10 further shows the active-



(a) The variance of traffic flow for both UTD19 and PEMS dataset.



(b) Histogram and Probability Density Function (PDF) of average MAPE on OSM+ and PEMS datasets.

Indicator	OSM+ (UTD19)	PEMS
Over 2nd	4.53	0.01
Over $2^{nd}(\%)$	5.87	0.01
Over 3rd	4.58	0.04
Over $3^{rd}(\%)$	5.94	0.02
Over worst	132.46	1.49
Over worst(%)	64.62	44.08

(c) Performance comparison on average MAE between OSM+(UTD19) dataset and PEMS dataset.

Figure 9. Comparison between the newly constructed OSM+(UTD19) dataset and PEMS dataset in three aspects.

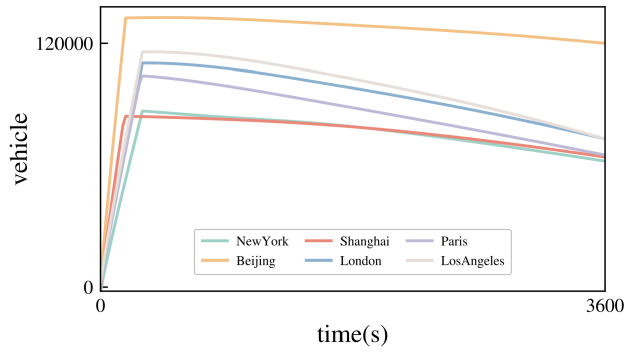


Figure 10. Active-vehicle curves of the six released traffic-control simulation scenarios under the generated demand profile.

vehicle curves of the six released scenarios under the generated demand profile. The curves describe how the number of vehicles simultaneously present in the simulator changes over time. All scenarios exhibit a rapid increase in active vehicles at the beginning of the simulation, followed by a gradual decrease as vehicles complete their trips. The OSM+ based scenarios involve substantially larger vehicle populations than the smaller previous benchmarks, demonstrating that the released environments can impose heavier traffic loads and better stress-test controller scalability. These results indicate that the generated demands are able to create non-trivial and diverse city-scale traffic-control settings.

5. Related Works

Map databases and services. Map databases and services are the base for various urban research. The major map databases include two types: commercial services and open-source maps. The commercial services are provided by major vendors like Google (Google Maps), Bing (Bing Maps), Apple (Apple Maps), Baidu (Baidu Maps), and Gaode (Gaode Maps). Though varying in serving regions, these map providers usually only provide high-level APIs (e.g., plotting, routing) for users to develop their own applications. These kind of APIs cannot support research need for querying external elements on the map, e.g., population count. To facilitate faster research progress and small business, some open-source map providers release their products, including OpenStreetMap (Haklay & Weber, 2008b; Bennett, 2010; Mooney et al., 2017), Mapbox (Eriksson & Rydkvist, 2015), Leaflet (Edler & Vetter, 2019) and GeoServer (Kshetri et al., 2021). These products store the map in various formats, which are difficult to be processed into a uniform format and need to be further cleaned for research use.

Map computing engine. To better utilize map data, many map computing engines have been proposed. These tools can be categorized into two types: commercial map tools and open-source map tools. For *commercial map engines*, Hu & Dai (2013) developed an online map application based on the Google Maps API, using commercial databases to provide users with complex data manipulation functions. Amani et al. (2020) processed remote sensing map data of Canadian agriculture on the Google Earth Engine (GEE) and obtained the annual crop list of Canada. Nguyen et al. (2019) combined GEE with the automatic water extraction index (AWEI) to address the long processing times in monitoring water surface dynamics. For *open-source map engines*, Elleuch et al. (2014) accomplished the process of converting large-scale databases collected by cars into road tracks. Boeing (2017) developed OSMnx, which simplifies data collection and road network analysis from the perspectives of graph theory, transportation, and urban design. Besides, existing OpenStreetMap data processing tools also include Osmosis (OSmosis), Nominatim (Nominatim) and OSRM (OSRM).

Table 5. Traffic-prediction results on 31 OSM+(UTD19) city datasets with seven baseline methods. Values are averaged over horizons 3, 6, and 12. MAE and MAPE(%) are reported; the best value in each row is bolded. OOM indicates that the run exceeded the 24GB GPU memory limit under our unified setting.

City	AGCRN		Crossformer		DCRNN		DLinear		FEDformer		GWNet		MTGNN	
	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE
AGB	47.92	44.03	**44.40**	**35.34**	OOM	OOM	47.80	37.84	51.94	45.19	47.06	37.14	46.97	40.47
BSL	64.51	55.38	62.82	65.68	119.15	184.45	61.50	**51.74**	**59.37**	61.78	81.07	106.88	78.41	93.81
BRN	51.00	231.93	**49.84**	**201.84**	OOM	OOM	52.18	253.27	55.58	248.00	50.59	319.09	70.90	405.07
BHX	112.08	70.94	**84.13**	**48.08**	303.46	195.15	111.22	66.91	119.52	65.15	107.09	66.83	91.68	49.44
BOL	**31.27**	21.07	32.74	**20.98**	38.00	26.82	37.27	34.12	37.87	29.83	35.03	25.67	32.31	21.12
BOD	71.65	39.69	67.13	**36.19**	232.07	276.51	**67.13**	44.54	70.14	46.29	74.18	57.18	89.14	56.70
BRE	**56.31**	36.52	58.08	**34.22**	OOM	OOM	63.27	42.47	61.42	41.98	57.01	36.98	56.69	35.50
KN	OOM	OOM	44.78	**48.85**	117.40	292.38	**38.69**	61.19	40.98	67.71	43.85	55.18	47.89	75.01
DA	57.22	51.76	**53.28**	50.75	OOM	OOM	54.76	53.99	57.41	61.16	54.69	51.74	57.20	**50.30**
ESS	41.95	**34.65**	40.47	41.68	174.64	294.77	50.35	43.85	44.70	41.95	38.99	34.87	**38.41**	36.46
FRA	163.16	54.95	145.88	47.61	179.03	51.37	**99.62**	**30.19**	107.78	31.76	190.85	62.23	284.10	92.89
GRZ	61.15	113.62	**52.78**	**66.74**	183.88	464.62	60.83	72.86	56.16	73.15	58.03	68.32	56.60	74.38
GRQ	69.64	35.57	68.02	**32.53**	158.26	114.53	**66.03**	37.54	79.09	42.20	67.99	34.95	74.99	39.00
HAM	46.50	44.89	44.49	44.87	97.85	108.12	46.69	49.81	47.85	50.69	**44.25**	**43.83**	45.02	44.18
INN	72.80	31.56	69.28	37.40	342.05	314.50	89.95	39.55	74.44	32.32	**67.03**	**28.53**	OOM	OOM
KS	81.26	106.06	86.38	118.43	233.90	427.98	75.29	107.43	89.83	127.22	**71.23**	**94.88**	191.45	316.68
MAN	106.16	42.54	97.48	41.10	336.42	280.35	101.38	46.21	110.81	52.15	**95.91**	**38.95**	97.30	40.74
MEL	50.24	45.88	**45.36**	42.73	OOM	OOM	63.72	66.55	53.25	56.25	51.91	**36.10**	45.48	40.26
RTM	**52.48**	**40.29**	53.83	50.52	179.76	347.68	68.83	53.19	68.43	65.17	67.03	50.91	57.34	41.07
SDR	103.63	59.74	102.25	65.34	262.60	271.61	97.97	54.70	125.51	95.71	**89.36**	47.38	97.54	**44.61**
SP	49.08	39.57	**47.93**	37.74	119.56	119.22	52.95	45.39	53.42	44.88	48.34	38.34	48.05	**37.48**
SXB	78.34	39.40	76.17	38.72	261.11	223.11	85.62	46.72	84.71	46.10	76.86	39.46	**76.01**	**37.36**
STR	58.93	20.37	56.60	19.52	68.19	23.30	65.80	24.52	68.38	23.48	**55.80**	**19.05**	OOM	OOM
TPE	136.50	48.04	134.51	48.18	502.95	274.25	142.61	46.21	149.12	53.31	**129.13**	**40.14**	130.36	41.42
TO	89.48	57.66	**81.70**	**44.44**	314.62	390.29	85.13	48.01	87.85	56.18	102.69	60.64	104.28	68.82
YTO	51.73	39.35	**51.54**	40.18	161.46	145.72	90.53	71.76	62.92	59.10	58.04	38.73	52.24	**37.42**
TLS	257.82	751.49	255.29	756.09	268.70	847.32	263.95	870.21	296.55	836.03	**255.26**	751.62	258.62	**730.09**
UTC	OOM	OOM	50.35	62.80	OOM	OOM	50.78	54.42	66.80	88.25	74.98	88.33	**39.92**	**36.74**
VNO	88.95	54.81	84.09	49.34	OOM	OOM	76.03	43.69	88.84	49.53	**73.80**	**39.27**	96.47	64.87
WOB	54.48	41.34	**52.21**	**39.71**	59.39	47.61	62.24	50.94	57.60	50.15	54.32	42.30	53.24	40.17
ZRH	OOM	OOM	54.73	36.93	OOM	OOM	60.36	43.84	60.12	43.74	66.51	53.31	**53.52**	**35.16**
# Win	3	2	10	10	0	0	4	2	1	0	9	8	4	8

Although these tools are able to process map data, they still suffer from two key limitations. First, the processed outputs are often not well-defined, uniform, or structured, and are typically tailored for one-time use. As a result, similar tasks require repetitive computation. Second, these tools usually run on local machines, limiting their capabilities to support large-scale analyses, such as global-level processing. The comparison of existing OpenStreetMap processing tools and OSM+ is shown in Table 6.

Table 6. Comparison of OSM data processing tools.

Method	Efficient Querying	Graph Structure Output
Overpass API	✗	✗
Osmosis	✗	✗
OSMnx	✗	✓
Nominatim	✓	✗
OSRM	✓	✗
OSM+	✓	✓

6. Conclusion

In this paper, we release OSM+ as a scalable data infrastructure for city-wide graph learning and urban computing, transforming raw OpenStreetMap records into a structured, worldwide road-network graph with cloud-supported query and conversion tools. By providing billion-level graph tables and standardized access utilities, OSM+ lowers the barrier to reproducible large-scale geospatial research and serves as a common spatial backbone for downstream studies, including functional city-boundary analysis, cross-city traffic forecasting, and large-scale traffic-control simulation. The new 31-city traffic prediction benchmark and six city-scale control scenarios reveal that existing models still face substantial challenges in robustness, memory efficiency, and scalability when moving from small curated datasets to realistic urban networks. We hope OSM+ will support future research and encourage more generalizable and scalable methods for understanding and optimizing cities.

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Impact Statement

Positive impact. OSM+ provides an open, graph-structured, billion-scale road-network resource and accompanying query/conversion utilities built from OpenStreetMap. It lowers the barrier for reproducible geospatial and transportation research: instead of repeatedly cleaning raw OSM exports and re-implementing fragile preprocessing pipelines, practitioners can directly access a unified road-graph representation, efficient spatial queries, and converters for tasks such as urban analytics, traffic forecasting, simulation-based policy evaluation, and multimodal geospatial model training.

Potential negative impact. Because OpenStreetMap is crowdsourced, OSM+ inherits regional variation in coverage and annotation quality. Models or benchmarks built on top of OSM+ may perform better in well-mapped regions and generalize poorly to sparsely mapped regions if this variation is ignored. In traffic forecasting, the same issue can also increase sensor-to-road matching noise when a sensor located on a missing minor road is projected to a more distant major road. Users should therefore report coverage, stratify evaluation by data quality when possible, preserve uncertainty in downstream analyses, and combine OSM+ with complementary local data sources for operational deployments. The UTD19-based traffic benchmark should be interpreted as a reproducible offline research benchmark rather than an online traffic-management ground truth.

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Appendix

A. Data and Code Access

Our OSM+ dataset is made available under the **Open Database License (ODbL 1.0)**: http://opendatacommons.org/licenses/odbl/1.0/. Any rights in individual contents of the database are licensed under the Database Contents License: http://opendatacommons.org/licenses/dbcl/1.0/. These tables are released as standard downloadable files through our official **Huggingface Dataset** (see detailed information in our official GitHub repository), allowing users to download complete tables or region-level shards for local processing.

Preprocessing pipeline. The complete data preprocessing pipeline is released at official GitHub repository. External users can follow the step-by-step tutorial to reproduce the data construction process from raw Geofabrik mirrors, including converting raw OSM files into global csv files and generating the corresponding OSM+ tables. All codes in this repository are released under **MIT License**.

Efficient query tools. We provide query tools based on the **City Brain Platform** to support efficient querying and management of OSM+ data. The operational workflows and scripts for using this platform with OSM+ are available at official GitHub repository. The query interface is useful for extracting city-scale subgraphs, issuing SQL-style spatial queries, and joining road segments with auxiliary tags without downloading all global-scale tables.

Benchmark code and data. All experimental configurations, source code, processed benchmark graphs, and converters for the downstream tasks in this paper are released at official GitHub repository. These resources support the reproduction of our benchmark experiments.

Third-party data boundary. The downstream benchmarks released in this paper integrate OSM+ with third-party upstream data sources such as UTD19 (Loder et al., 2020) and nightlight data (Weier, 2003). These third-party sources remain subject to their original licenses, access policies, and redistribution terms. We release the alignment code and processed benchmark structures to support reproducibility, but we do not claim ownership of third-party upstream data.

B. Dataset Maintenance

The OSM+ release evaluated in this paper is a static snapshot processed from Geofabrik OpenStreetMap data downloaded in **December 2022**, and this fixed snapshot is used for all reported experiments. The dataset statistics reported in Table 1 are computed from the updated OSM+ release based on the **March 2026** OpenStreetMap snapshot, so as to reflect the current scale of the released dataset. Since OpenStreetMap is continuously edited by volunteers, a single static copy can become stale over time; therefore, we maintain OSM+ using a time-based versioning strategy and plan to update the dataset every three months. Each release will be named by its release month, e.g., OSM+vYYYY.MM, and archived independently so that researchers can reproduce a specific benchmark snapshot while also accessing newer road-network states for current applications. In addition, the released preprocessing pipeline allows users to process a newer Geofabrik .osm.pbf file for a particular region of interest without waiting for the next official global release.

C. Preprocessing Pipeline

The preprocessing scripts and tutorials of OSM+ are available at official GitHub repository. The overall procedure of the OpenStreetMap data processing can be described as the following steps:

1. Download continental OSM files from http://download.geofabrik.de/.
2. Decompose continental OSM files (larger than 100GB) into several smaller OSM files.
3. Convert OSM files into CSV files, including **nodes.csv** and **edges.csv**. In node files, each record represents a node on OpenStreetMap with about 30 attributes. In edge files, each record represents an edge-like OSM object.
4. Split linestrings on OpenStreetMap and transform **edges.csv** files into start–end pairs. Each edge file is transformed into a **fulltag.csv** file and a **split.csv** file. The schema of **fulltag.csv** is the same as **edges.csv** except for removing the "nodes" tag. Each record in **split.csv** represents a link on OpenStreetMap.
5. Upload these files to MaxCompute/ODPS for distributed cleaning and joining.

6. Select nodes and edges on the roadnet from tables using cloud-computing tools. Finally, three roadnet-only tables and three full tables are generated. Detailed schemas are introduced in our official Huggingface Dataset.

D. Auxiliary Attribute Completeness

OSM+ parses auxiliary OSM tags, including road hierarchy and POI-related attributes, into structured fields. These attributes are useful for downstream tasks, but their completeness naturally varies across regions because OSM is maintained by local contributors. Well-mapped regions often provide dense POIs and consistent road hierarchy tags, whereas sparsely mapped regions may contain many generic labels such as `unclassified` and fewer POI annotations. We preserve this original distribution rather than interpolating missing attributes. Users who rely on auxiliary attributes should report attribute coverage, such as the fraction of road edges with valid hierarchy tags and POI density per square kilometer, before drawing cross-region conclusions.

E. Detailed Experiment Settings

E.1. Basic Query Operation

This section describes the details of optimizing basic query operations. For point query, we divide the entire map into grids of 0.2° by 0.2° in latitude and longitude, and determine which grid the point to be queried is located in. Then, we only need to retrieve data points in adjacent grids during retrieval. For the nearest neighbor query, we also used the "window check" method. In addition, we utilized the k-d tree, a binary tree that represents a division of k-dimensional space to reduce the time complexity.

E.2. City Boundary Detection

This section mainly describes the process of using OSM+ data for city boundary detection. First, we select several representative areas for experiment, including the China Yangtze River Delta (115°E∼122°E, 29°N∼32°N), New York in the USA (84°W∼70.5°W, 37.8°N∼45.6°N), Central Europe (2°W∼11.5°E, 45.5°N∼53°N), and Nigeria in Africa (0.5°E∼14°E, 4.2°N∼11.7°N). We integrate the roadnet data, raster image population data, and raster image nightlight data of these areas by converting them into point-wise data. Next, we perform clustering on the road net data, nightlight data, and population data respectively, and then weigh and sum the results of clustering to obtain a new clustering result. After that, we divide the entire map into a grid structure with a size of 0.05° in latitude and longitude. We then count the number of points of each type within each grid and assign the grid to the category with the most points, obtaining a rough boundary. Finally, we examine the empty grids by considering a grid nine times its size centered on it and reclassifying it to make the boundary smoother. This process allows us to discover a more reasonable city boundary.

E.3. City-scale Traffic Prediction

We conduct all the experiments on machines with two NVIDIA RTX 3090 GPUs (24GB VRAM each) and 128 GB of CPU memory on Ubuntu 20.04. All models are implemented in Python 3. For the problem setting, both the input and output sequence lengths are set to 12. We use traffic-flow data from UTD19 (Loder et al., 2020) to construct a reproducible offline benchmark, rather than an online deployment setting or an authoritative ground truth for operational traffic management. For all 31 city datasets, the training, validation, and test sets are split at a ratio of 6:2:2. We evaluate performance using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

Sensor-to-road matching. For each UTD19 city, we first extract the corresponding OSM+ road-network subgraph by spatial range. We then align UTD19 sensor or measurement coordinates to OSM+ through nearest-edge matching: each observation location is projected to the closest physical road segment in the OSM+ roadnet, and the associated road-network topology is used to construct the prediction graph. This matching procedure is deterministic and scalable, but it may introduce localization drift when OpenStreetMap coverage is sparse or when the true minor road is missing. A global quantitative matching-error audit would require lane-level ground truth or high-frequency trajectories for all cities, which are not available across the full benchmark. We therefore preserve the matching noise and report it as part of the realistic cross-city benchmark challenge.

Here we list some details of our implemented baseline methods. For hyperparameters that are not mentioned, we adopt the default hyperparameters from BasicTS (Shao et al., 2023):

AGCRN (Bai et al., 2020). We use two layers of AGCRN to capture the node-specific spatial and temporal dynamics. For the hyperparameters, we set the hidden unit to 64 for all the AGCRN cells and the batch size to 64. We set the learning rate to 0.003 and the embedding dimension to 64 for all 31 city datasets extracted from the OSM+ dataset. Besides, we choose L_1 Loss as the loss function.

Crossformer (Zhang & Yan, 2022). When implementing Crossformer, we set segment length L_{seg} to 24, as it is related to both the model performance and computation efficiency. Besides, we set the window size to 2. We use Adam optimizer with a 0.0002 learning rate and 0.0005 weight decay rate.

DCRNN (Li et al., 2017). Both encoder and decoder contain two recurrent layers. In each recurrent layer, there are 64 units and the initial learning rate is set to 0.003. Besides, the maximum step of random walks, i.e., K, is set to 2. For scheduled sampling, the thresholded inverse sigmoid function is used as the probability decay:

$$\epsilon_i = \frac{\tau}{\tau + \exp(i/\tau)} \tag{1}$$

where i is the number of iterations while τ is the parameter to control the speed of convergence, which is set to 2,000 in the experiments.

DLinear (Diagne et al., 2012). For implementation details about DLinear, we adopt the default hyperparameters from (Shao et al., 2023) to train the models. The training epoch is set to 100.

FEDformer (Zhou et al., 2022). The FEDformer is trained using the Adam optimizer with a learning rate of 0.0005. The batch size is set to 64. An early stopping counter is employed to stop the training process after three epochs if no loss degradation on the validation set is observed.

GWNet (Wu et al., 2019). We use two layers of Graph WaveNet with a sequence of dilation factors $\{1, 2\}$. We randomly initialize node embeddings by a uniform distribution with a size of 10. We train our model using the Adam optimizer with an initial learning rate of 0.0005. Dropout with $p = 0.3$ is applied to the outputs of the graph convolution layer.

E.4. Traffic Policy Control

We use the script provided by CBData (Liang et al., 2023) to convert the OSM+ roadnet file to the corresponding format required by CBEngine. We then conduct traffic-flow simulation experiments in six cities (Beijing, London, Los Angeles, New York, Paris, and Shanghai). The simulation runs for 3600 steps, with traffic demand injected uniformly during the first 300 steps. The released scenario files include the converted road network, signal phase configuration, generated demand, and scripts for computing standard control metrics such as average travel time, queue length, and delay.

The released scenarios are designed to support comparable controller evaluation. Fixed-time, max-pressure, and learning-based controllers can be evaluated using the same roadnet, demand files, and metric scripts, ensuring that future methods can be compared using identical city-scale inputs.

F. Extended Experiment Results

To further illustrate the supporting performance of the OSM+ dataset over traffic flow prediction task, we repeat each set of experiments five times and report their means and standard deviations on horizon 3, 6 and 12, following the common setting in this problem (Tedjopurnomo et al., 2020). Comprehensive results are shown in Table 7, Table 8 and Table 9. The correspondence between city names and their abbreviations is shown as follows: AGB (Augsburg), BSL (Basel), BRN (Bern), BHX (Birmingham), BOL (Bolton), BOD (Bordeaux), BRE (Bremen), KN (Constance), DA (Darmstadt), ESS (Essen), FRA (Frankfurt), GRZ (Graz), GRQ (Groningen), HAM (Hamburg), INN (Innsbruck), KS (Kassel), MAN (Manchester), MEL (Melbourne), RTM (Rotterdam), SDR (Santander), SP (Speyer), SXB (Strasbourg), STR (Stuttgart), TPE (Taipeh/Taipei), TO (Torino), YTO (Toronto), TLS (Toulouse), UTC (Utrecht), VNO (Vilnius), WOB (Wolfsburg), and ZRH (Zurich).

Table 7. The performance comparison for seven baseline methods over 31 real-world city datasets with **horizon=3**. The best results in each row are in bold. All experiments are repeated five times, and the mean and standard deviation are reported.

City	Metric	AGCRN	Crossformer	DCRNN	DLinear	FEDformer	GWNet	MTGNN

AGB	MAE	42.53 ± 0.22	**39.89 ± 0.11**	OOM	40.56 ± 0.00	46.15 ± 0.03	40.65 ± 0.19	40.76 ± 0.38
	RMSE	87.82 ± 0.22	85.21 ± 0.92	OOM	82.00 ± 0.00	99.68 ± 0.34	**81.14 ± 1.17**	83.04 ± 0.33
	MAPE(%)	38.97 ± 0.18	**31.21 ± 1.72**	OOM	32.43 ± 0.02	39.76 ± 0.10	32.09 ± 0.95	33.27 ± 2.39
BSL	MAE	55.00 ± 0.15	49.76 ± 0.96	116.48 ± 1.50	**49.12 ± 0.00**	52.55 ± 0.18	68.47 ± 0.15	65.86 ± 0.60
	RMSE	82.77 ± 0.08	**74.24 ± 2.83**	155.13 ± 0.43	79.39 ± 0.03	76.74 ± 0.37	93.83 ± 1.64	90.52 ± 1.20
	MAPE(%)	45.85 ± 0.35	**43.38 ± 2.60**	171.29 ± 7.33	42.93 ± 0.01	52.67 ± 1.15	91.08 ± 3.34	75.48 ± 2.24
BRN	MAE	46.82 ± 1.30	**45.65 ± 0.17**	OOM	48.76 ± 0.02	51.16 ± 0.10	47.49 ± 0.56	59.36 ± 0.86
	RMSE	468.78 ± 9.95	**454.56 ± 1.64**	OOM	454.98 ± 0.18	431.93 ± 0.45	440.49 ± 7.76	477.32 ± 13.68
	MAPE(%)	**222.50 ± 13.23**	233.32 ± 18.35	OOM	229.36 ± 2.05	232.49 ± 17.64	278.14 ± 19.92	347.74 ± 20.37
BHX	MAE	88.84 ± 0.35	84.86 ± 7.57	289.70 ± 11.26	92.86 ± 0.80	104.66 ± 0.44	90.95 ± 0.18	**84.13 ± 0.64**
	RMSE	137.05 ± 0.61	134.05 ± 8.50	425.16 ± 37.43	144.99 ± 0.88	158.84 ± 0.19	144.59 ± 1.12	**130.56 ± 0.88**
	MAPE(%)	43.38 ± 0.35	**40.51 ± 2.58**	191.60 ± 57.29	48.95 ± 0.11	52.14 ± 0.90	51.86 ± 1.83	45.67 ± 3.76
BOL	MAE	**29.63 ± 0.27**	32.03 ± 1.73	32.74 ± 0.38	33.31 ± 0.02	34.19 ± 0.04	32.69 ± 1.92	29.68 ± 0.23
	RMSE	84.46 ± 0.07	84.23 ± 0.25	87.89 ± 0.04	83.81 ± 0.06	87.51 ± 0.13	84.37 ± 0.56	**81.28 ± 0.13**
	MAPE(%)	**18.44 ± 1.09**	19.69 ± 0.44	22.48 ± 0.69	30.71 ± 0.10	26.86 ± 1.25	22.51 ± 0.53	18.63 ± 0.50
BOD	MAE	62.34 ± 0.22	60.21 ± 0.60	244.58 ± 20.30	**58.23 ± 0.00**	58.51 ± 0.30	59.78 ± 0.05	71.17 ± 1.11
	RMSE	100.52 ± 0.47	95.41 ± 0.16	324.45 ± 24.77	92.86 ± 0.01	**92.09 ± 0.27**	93.00 ± 0.09	109.95 ± 2.32
	MAPE(%)	37.45 ± 1.21	**33.47 ± 0.43**	280.19 ± 14.03	37.59 ± 0.05	38.28 ± 0.84	43.06 ± 0.33	45.75 ± 0.42
BRE	MAE	55.40 ± 0.02	56.33 ± 1.40	OOM	58.99 ± 0.03	58.80 ± 0.04	55.45 ± 0.09	**55.18 ± 0.17**
	RMSE	92.18 ± 0.08	93.06 ± 1.74	OOM	97.06 ± 0.11	95.33 ± 0.02	**91.31 ± 0.11**	91.72 ± 0.48
	MAPE(%)	36.08 ± 0.26	**33.69 ± 1.66**	OOM	40.12 ± 0.12	40.41 ± 0.49	36.10 ± 0.27	34.93 ± 0.22
KN	MAE	OOM	39.17 ± 1.36	119.00 ± 0.09	**35.47 ± 0.00**	38.35 ± 0.37	37.55 ± 0.55	40.06 ± 0.38
	RMSE	OOM	62.75 ± 2.15	152.84 ± 0.67	**54.99 ± 0.00**	58.78 ± 0.32	58.21 ± 0.67	61.15 ± 0.86
	MAPE(%)	OOM	47.64 ± 0.56	293.93 ± 0.88	54.02 ± 0.17	61.77 ± 1.36	**47.54 ± 0.59**	58.71 ± 1.08
DA	MAE	56.32 ± 0.04	51.12 ± 0.79	OOM	**50.77 ± 0.00**	54.26 ± 0.08	51.80 ± 0.11	54.58 ± 0.23
	RMSE	88.65 ± 0.20	77.01 ± 1.12	OOM	**75.41 ± 0.00**	79.78 ± 0.25	78.39 ± 0.16	86.54 ± 1.08
	MAPE(%)	50.79 ± 0.00	**45.51 ± 0.49**	OOM	51.38 ± 0.02	58.16 ± 0.89	48.27 ± 0.64	48.10 ± 0.11
ESS	MAE	40.66 ± 0.28	36.95 ± 0.10	172.66 ± 0.19	41.06 ± 0.00	40.51 ± 0.07	36.86 ± 0.08	**35.94 ± 0.07**
	RMSE	59.07 ± 0.76	53.29 ± 0.27	224.47 ± 3.72	59.39 ± 0.01	58.41 ± 0.03	53.72 ± 0.22	**52.57 ± 0.03**
	MAPE(%)	35.54 ± 1.10	34.79 ± 1.28	302.81 ± 15.18	34.62 ± 0.40	36.01 ± 0.71	32.85 ± 0.28	**31.51 ± 2.29**
FRA	MAE	124.69 ± 0.05	140.18 ± 14.57	185.87 ± 2.29	**71.58 ± 0.30**	84.83 ± 1.13	139.06 ± 1.84	170.44 ± 12.29
	RMSE	157.09 ± 0.09	166.53 ± 16.07	237.20 ± 2.82	**90.90 ± 1.09**	108.23 ± 2.00	163.25 ± 2.09	190.32 ± 12.38
	MAPE(%)	39.40 ± 0.02	41.18 ± 3.19	47.65 ± 0.59	**20.21 ± 0.07**	23.41 ± 0.15	42.48 ± 0.68	51.70 ± 3.62
GRZ	MAE	59.06 ± 0.02	**50.55 ± 1.02**	174.96 ± 0.00	52.27 ± 0.00	50.95 ± 0.24	55.07 ± 0.74	53.48 ± 0.62
	RMSE	89.07 ± 0.02	74.28 ± 0.84	220.49 ± 0.00	75.95 ± 0.01	**74.23 ± 0.17**	81.10 ± 1.00	82.35 ± 1.49
	MAPE(%)	116.57 ± 0.80	68.30 ± 4.29	467.17 ± 0.00	**63.68 ± 0.15**	66.96 ± 2.12	68.11 ± 1.24	71.41 ± 0.40
GRQ	MAE	63.24 ± 0.01	62.25 ± 1.53	162.87 ± 0.00	**60.08 ± 0.41**	72.11 ± 1.47	64.33 ± 0.16	63.21 ± 1.50
	RMSE	87.02 ± 0.08	86.03 ± 0.07	218.69 ± 0.00	**82.72 ± 0.56**	99.96 ± 1.73	88.41 ± 0.03	86.27 ± 1.51
	MAPE(%)	30.46 ± 0.19	**28.82 ± 0.24**	112.24 ± 0.00	31.41 ± 0.29	36.83 ± 0.57	31.79 ± 0.19	30.87 ± 1.08
HAM	MAE	45.68 ± 0.09	43.50 ± 0.10	97.51 ± 0.56	44.82 ± 0.00	46.34 ± 0.05	**43.28 ± 0.01**	44.05 ± 0.09
	RMSE	75.36 ± 0.60	70.76 ± 0.19	150.67 ± 2.84	73.13 ± 0.01	74.96 ± 0.18	**70.69 ± 0.07**	73.02 ± 0.19
	MAPE(%)	44.67 ± 0.36	**42.79 ± 0.51**	111.42 ± 5.09	47.93 ± 0.10	48.93 ± 0.17	43.48 ± 0.16	43.30 ± 0.13
INN	MAE	70.05 ± 0.15	67.34 ± 1.24	333.14 ± 2.31	76.43 ± 0.01	70.66 ± 0.65	**65.65 ± 0.12**	OOM
	RMSE	101.84 ± 0.05	97.03 ± 0.57	443.91 ± 2.79	113.93 ± 0.09	102.62 ± 0.80	**95.28 ± 0.17**	OOM
	MAPE(%)	31.38 ± 0.83	35.24 ± 5.44	304.36 ± 0.76	33.84 ± 0.45	30.70 ± 0.10	**28.00 ± 0.42**	OOM

KS	MAE	69.88 ± 0.37	73.22 ± 1.27	244.28 ± 0.00	**63.22 ± 2.52**	80.77 ± 0.65	63.34 ± 1.06	154.46 ± 1.90
	RMSE	203.93 ± 0.33	204.17 ± 2.19	342.80 ± 0.00	158.72 ± 4.21	177.02 ± 0.31	**155.08 ± 0.53**	233.89 ± 3.73
	MAPE(%)	**80.45 ± 1.04**	87.42 ± 3.53	440.12 ± 0.00	81.40 ± 4.85	105.62 ± 1.65	81.47 ± 2.36	235.91 ± 0.42
MAN	MAE	97.96 ± 0.71	87.80 ± 0.65	336.72 ± 0.00	92.36 ± 5.65	99.05 ± 1.29	**84.26 ± 0.20**	85.24 ± 0.93
	RMSE	169.67 ± 0.73	160.48 ± 1.62	448.89 ± 0.00	156.98 ± 7.85	167.30 ± 2.46	**151.48 ± 0.49**	154.15 ± 2.57
	MAPE(%)	39.66 ± 1.35	37.73 ± 0.54	283.06 ± 0.00	42.24 ± 2.20	46.21 ± 1.60	**32.83 ± 0.26**	36.21 ± 1.39
MEL	MAE	37.39 ± 0.00	36.21 ± 0.95	OOM	42.57 ± 0.85	40.80 ± 0.58	36.26 ± 0.01	**35.86 ± 0.95**
	RMSE	56.63 ± 0.01	**54.30 ± 2.21**	OOM	64.49 ± 1.03	60.80 ± 1.36	54.35 ± 0.06	54.78 ± 1.68
	MAPE(%)	37.60 ± 0.03	35.63 ± 0.72	OOM	37.82 ± 0.99	43.56 ± 0.04	**27.22 ± 0.07**	31.77 ± 0.80
RTM	MAE	**49.08 ± 0.16**	50.46 ± 0.06	170.03 ± 0.00	55.93 ± 0.01	58.45 ± 0.73	54.45 ± 0.28	51.45 ± 0.08
	RMSE	**87.20 ± 0.12**	87.80 ± 0.13	232.29 ± 0.00	95.05 ± 0.02	96.26 ± 0.90	92.68 ± 0.52	88.92 ± 0.36
	MAPE(%)	**36.76 ± 0.05**	41.26 ± 0.01	320.02 ± 0.00	44.61 ± 0.19	57.44 ± 1.81	43.72 ± 1.04	38.46 ± 1.26
SDR	MAE	88.07 ± 0.25	86.61 ± 1.76	259.05 ± 0.00	80.68 ± 0.02	107.62 ± 1.33	**77.84 ± 0.41**	78.50 ± 2.26
	RMSE	230.51 ± 0.94	231.19 ± 0.49	434.97 ± 0.00	**187.89 ± 0.01**	216.83 ± 2.05	202.47 ± 1.86	211.89 ± 0.85
	MAPE(%)	54.00 ± 1.15	53.10 ± 6.85	257.82 ± 0.00	43.89 ± 0.16	81.61 ± 0.53	44.22 ± 3.89	**36.34 ± 0.17**
SP	MAE	48.92 ± 0.05	**47.51 ± 0.03**	121.12 ± 2.04	50.74 ± 0.01	52.37 ± 0.11	47.84 ± 0.05	47.75 ± 0.11
	RMSE	70.45 ± 0.06	**68.34 ± 0.07**	172.55 ± 9.56	72.60 ± 0.06	74.68 ± 0.06	68.91 ± 0.17	68.90 ± 0.02
	MAPE(%)	39.84 ± 0.53	**36.77 ± 0.11**	102.07 ± 25.06	43.49 ± 0.00	43.79 ± 0.59	37.92 ± 0.33	37.04 ± 0.13
SXB	MAE	76.69 ± 0.07	73.93 ± 0.14	259.67 ± 0.00	78.84 ± 0.05	80.54 ± 0.24	74.23 ± 0.03	**73.87 ± 0.41**
	RMSE	134.85 ± 0.14	**130.71 ± 0.32**	360.44 ± 0.00	138.48 ± 0.03	141.88 ± 0.15	131.69 ± 0.05	131.14 ± 0.23
	MAPE(%)	40.04 ± 1.09	40.73 ± 2.09	223.03 ± 0.00	43.50 ± 0.18	43.51 ± 0.74	37.64 ± 0.26	**36.53 ± 0.63**
STR	MAE	57.40 ± 0.43	57.84 ± 3.60	59.86 ± 1.67	60.07 ± 0.08	68.33 ± 1.18	**55.70 ± 0.02**	OOM
	RMSE	74.79 ± 0.28	75.55 ± 4.24	78.25 ± 3.02	78.53 ± 0.16	89.06 ± 1.27	**72.13 ± 0.07**	OOM
	MAPE(%)	18.44 ± 0.08	19.11 ± 2.18	19.07 ± 0.70	20.01 ± 0.01	22.32 ± 0.45	**18.10 ± 0.06**	OOM
TPE	MAE	126.81 ± 0.07	126.02 ± 3.81	490.25 ± 8.98	125.41 ± 0.01	134.61 ± 0.09	**117.30 ± 0.40**	121.00 ± 0.90
	RMSE	555.59 ± 1.23	561.67 ± 10.18	988.24 ± 11.03	**482.59 ± 0.38**	540.48 ± 5.15	493.02 ± 0.48	512.90 ± 1.36
	MAPE(%)	42.20 ± 0.35	45.74 ± 5.09	266.56 ± 4.39	41.80 ± 0.12	47.72 ± 0.55	**38.30 ± 1.75**	40.01 ± 1.54
TO	MAE	77.75 ± 0.14	71.06 ± 0.96	313.88 ± 0.21	**68.52 ± 0.01**	74.48 ± 0.07	74.91 ± 1.05	80.07 ± 1.05
	RMSE	131.86 ± 0.13	114.97 ± 2.10	415.51 ± 1.03	**111.00 ± 0.07**	118.37 ± 0.35	116.63 ± 0.82	124.72 ± 1.31
	MAPE(%)	50.86 ± 0.32	**39.42 ± 0.29**	400.19 ± 7.19	40.27 ± 0.47	48.37 ± 0.98	46.33 ± 2.97	52.21 ± 2.04
YTO	MAE	45.81 ± 0.12	45.25 ± 0.40	129.55 ± 18.89	59.49 ± 0.01	52.16 ± 0.04	46.68 ± 0.25	**44.65 ± 0.07**
	RMSE	77.29 ± 0.30	**74.48 ± 0.06**	192.41 ± 21.36	95.84 ± 0.09	81.85 ± 0.04	77.47 ± 0.47	75.03 ± 0.00
	MAPE(%)	33.49 ± 0.02	**32.72 ± 1.55**	114.20 ± 27.77	42.70 ± 0.66	46.45 ± 1.69	35.18 ± 0.38	33.36 ± 2.07
TLS	MAE	257.32 ± 0.36	**255.21 ± 0.28**	264.24 ± 6.15	263.84 ± 0.00	296.07 ± 0.02	255.32 ± 0.00	259.00 ± 0.03
	RMSE	349.78 ± 0.86	342.51 ± 0.43	352.70 ± 9.62	348.15 ± 0.03	410.70 ± 0.70	**341.22 ± 0.34**	349.09 ± 0.86
	MAPE(%)	754.88 ± 3.75	746.38 ± 4.66	791.79 ± 77.77	872.83 ± 0.91	836.22 ± 1.06	747.55 ± 5.53	**726.09 ± 9.80**
UTC	MAE	OOM	51.67 ± 0.80	OOM	44.40 ± 1.21	61.74 ± 1.37	75.85 ± 0.25	**39.61 ± 0.20**
	RMSE	OOM	83.26 ± 3.39	OOM	75.24 ± 1.49	91.06 ± 0.40	120.97 ± 0.66	**68.41 ± 0.37**
	MAPE(%)	OOM	57.21 ± 15.30	OOM	45.37 ± 1.44	77.81 ± 4.77	91.81 ± 5.44	**37.55 ± 2.65**
VNO	MAE	82.75 ± 0.08	77.98 ± 0.27	OOM	**69.08 ± 1.53**	81.57 ± 0.28	69.25 ± 0.23	89.80 ± 1.00
	RMSE	112.24 ± 0.29	106.39 ± 0.56	OOM	**94.18 ± 2.13**	110.65 ± 0.82	94.42 ± 0.06	118.34 ± 1.00
	MAPE(%)	47.89 ± 0.26	42.38 ± 0.14	OOM	35.59 ± 0.75	42.38 ± 0.60	**35.32 ± 0.81**	57.14 ± 1.75
WOB	MAE	52.66 ± 0.05	51.73 ± 1.69	54.51 ± 0.05	56.05 ± 0.02	54.89 ± 0.01	51.27 ± 0.18	**51.12 ± 0.01**
	RMSE	81.81 ± 0.31	79.60 ± 3.58	84.03 ± 0.30	85.65 ± 0.02	82.31 ± 0.02	**78.15 ± 0.24**	78.27 ± 0.04
	MAPE(%)	40.76 ± 0.38	**37.38 ± 0.17**	42.23 ± 0.24	46.71 ± 0.06	46.64 ± 0.10	39.56 ± 0.58	39.41 ± 0.80

ZRH	MAE	OOM	53.23 ± 0.05	OOM	56.33 ± 0.00	56.72 ± 0.08	59.24 ± 7.75	**52.34 ± 0.18**
	RMSE	OOM	75.13 ± 0.05	OOM	79.32 ± 0.01	79.36 ± 0.01	83.96 ± 11.26	**74.05 ± 0.34**
	MAPE(%)	OOM	35.77 ± 1.18	OOM	40.87 ± 0.04	41.31 ± 0.53	44.88 ± 11.14	**34.95 ± 0.46**

Table 8. The performance comparison for seven baseline methods over 31 real-world city datasets with **horizon=6**. The best results in each row are in bold. All experiments are repeated five times, and the mean and standard deviation are reported.

City	Metric	AGCRN	Crossformer	DCRNN	DLinear	FEDformer	GWNet	MTGNN
AGB	MAE	47.27 ± 0.41	**43.29 ± 0.11**	OOM	46.25 ± 0.00	50.21 ± 0.02	46.22 ± 0.59	45.61 ± 0.55
	RMSE	100.54 ± 0.22	**95.50 ± 1.21**	OOM	97.90 ± 0.03	112.57 ± 0.37	97.17 ± 2.27	97.30 ± 0.43
	MAPE(%)	43.35 ± 0.73	**33.68 ± 0.45**	OOM	36.33 ± 0.15	42.69 ± 0.09	36.93 ± 1.56	37.46 ± 3.09
BSL	MAE	63.95 ± 0.02	61.81 ± 3.57	118.10 ± 0.84	**59.01 ± 0.01**	59.94 ± 1.42	81.55 ± 1.76	78.88 ± 1.13
	RMSE	94.86 ± 0.09	91.78 ± 4.55	157.37 ± 1.12	99.52 ± 0.04	**87.06 ± 2.86**	114.99 ± 4.41	110.81 ± 2.49
	MAPE(%)	54.81 ± 0.20	68.11 ± 2.43	175.12 ± 9.70	**49.76 ± 0.16**	59.65 ± 0.08	109.78 ± 3.98	91.64 ± 2.08
BRN	MAE	49.20 ± 1.52	**48.47 ± 0.32**	OOM	51.65 ± 0.01	54.01 ± 0.18	50.27 ± 0.10	69.27 ± 0.82
	RMSE	481.50 ± 16.19	465.00 ± 1.10	OOM	464.66 ± 0.33	**450.70 ± 0.37**	468.88 ± 0.39	508.09 ± 12.22
	MAPE(%)	216.94 ± 6.34	**186.02 ± 20.85**	OOM	244.39 ± 5.93	242.43 ± 19.54	293.55 ± 20.22	411.51 ± 30.03
BHX	MAE	101.87 ± 0.00	**86.82 ± 7.11**	299.18 ± 30.69	103.14 ± 2.28	113.03 ± 0.16	99.17 ± 0.39	89.68 ± 0.34
	RMSE	148.33 ± 0.23	**133.22 ± 9.67**	426.46 ± 61.57	156.29 ± 2.41	175.26 ± 0.51	150.67 ± 0.76	137.11 ± 1.22
	MAPE(%)	59.01 ± 0.56	**45.71 ± 1.30**	229.60 ± 44.64	57.28 ± 1.81	57.12 ± 0.60	56.87 ± 5.04	49.85 ± 1.96
BOL	MAE	**30.99 ± 0.09**	32.53 ± 1.31	37.09 ± 0.61	35.99 ± 0.07	35.71 ± 0.38	33.47 ± 0.60	31.57 ± 0.32
	RMSE	87.74 ± 0.14	87.00 ± 0.68	93.43 ± 0.01	88.23 ± 0.03	87.93 ± 0.81	82.40 ± 0.12	**82.26 ± 0.01**
	MAPE(%)	**19.58 ± 1.26**	20.30 ± 0.26	26.38 ± 0.99	33.56 ± 0.24	29.66 ± 1.49	28.79 ± 0.72	21.63 ± 0.37
BOD	MAE	69.25 ± 0.17	65.32 ± 0.92	242.05 ± 12.05	**64.43 ± 0.01**	64.68 ± 0.20	70.55 ± 0.22	86.45 ± 0.91
	RMSE	112.27 ± 0.50	103.95 ± 0.50	321.50 ± 16.12	102.17 ± 0.01	**101.21 ± 0.16**	108.43 ± 0.09	135.49 ± 2.53
	MAPE(%)	39.52 ± 1.05	**35.61 ± 0.49**	280.80 ± 1.26	42.14 ± 0.23	42.19 ± 0.72	52.86 ± 0.88	55.08 ± 1.37
BRE	MAE	56.30 ± 0.01	56.94 ± 0.75	OOM	62.32 ± 0.01	60.38 ± 0.08	56.86 ± 0.01	**56.26 ± 0.27**
	RMSE	93.86 ± 0.02	94.36 ± 0.99	OOM	101.83 ± 0.07	97.83 ± 0.11	93.62 ± 0.02	**93.58 ± 0.67**
	MAPE(%)	36.30 ± 0.14	**34.78 ± 0.07**	OOM	41.95 ± 0.28	41.36 ± 0.48	37.02 ± 0.02	35.45 ± 0.29
KN	MAE	OOM	43.36 ± 1.67	118.57 ± 2.65	**37.85 ± 0.00**	40.09 ± 0.24	43.01 ± 1.18	45.06 ± 0.30
	RMSE	OOM	69.19 ± 2.16	151.98 ± 2.99	61.54 ± 0.01	**61.47 ± 0.14**	68.27 ± 1.68	68.58 ± 0.14
	MAPE(%)	OOM	**47.64 ± 0.45**	294.00 ± 8.92	59.41 ± 0.17	64.05 ± 1.65	53.57 ± 1.58	68.56 ± 3.07
DA	MAE	58.34 ± 0.14	**53.03 ± 0.05**	OOM	53.93 ± 0.01	56.35 ± 0.07	55.08 ± 0.19	59.16 ± 0.15
	RMSE	92.05 ± 0.44	**80.86 ± 0.21**	OOM	81.10 ± 0.07	83.79 ± 0.23	84.75 ± 0.59	96.79 ± 0.57
	MAPE(%)	51.28 ± 0.03	**47.58 ± 0.26**	OOM	53.46 ± 0.08	59.83 ± 0.81	50.41 ± 1.14	50.52 ± 0.23
ESS	MAE	41.58 ± 0.11	39.69 ± 0.44	173.69 ± 0.75	47.27 ± 0.00	43.48 ± 0.01	38.23 ± 0.14	**37.59 ± 0.06**
	RMSE	61.02 ± 0.33	57.73 ± 0.93	226.42 ± 2.28	68.86 ± 0.02	63.05 ± 0.26	56.05 ± 0.23	**55.76 ± 0.09**
	MAPE(%)	34.29 ± 0.34	40.50 ± 4.51	299.82 ± 15.01	40.99 ± 0.15	39.62 ± 0.40	34.16 ± 0.17	**34.01 ± 2.44**
FRA	MAE	158.46 ± 0.09	141.89 ± 33.64	187.73 ± 7.77	**93.49 ± 1.42**	107.62 ± 2.29	173.62 ± 3.08	258.38 ± 13.05
	RMSE	192.87 ± 0.02	172.02 ± 35.30	236.87 ± 10.11	**115.65 ± 1.17**	136.75 ± 3.69	202.44 ± 2.63	279.20 ± 13.65
	MAPE(%)	52.05 ± 0.01	43.28 ± 8.91	52.30 ± 1.53	**27.45 ± 0.40**	30.92 ± 0.64	54.62 ± 1.16	81.61 ± 3.79
GRZ	MAE	60.79 ± 0.01	**52.70 ± 0.96**	185.12 ± 0.00	58.29 ± 0.01	54.03 ± 0.24	58.32 ± 0.87	56.03 ± 0.20
	RMSE	91.77 ± 0.08	**77.53 ± 0.71**	233.72 ± 0.00	84.30 ± 0.04	78.69 ± 0.12	86.47 ± 1.17	87.69 ± 0.61
	MAPE(%)	110.69 ± 1.77	**65.44 ± 6.48**	465.95 ± 0.00	72.50 ± 1.27	69.79 ± 3.62	71.75 ± 1.36	72.35 ± 0.12
GRQ	MAE	68.43 ± 0.01	65.61 ± 1.26	161.01 ± 0.00	**64.78 ± 0.91**	79.46 ± 0.32	66.49 ± 0.22	71.04 ± 3.46

	RMSE	93.25 ± 0.24	90.55 ± 0.25	217.62 ± 0.00	**89.18 ± 1.54**	110.72 ± 0.49	91.41 ± 0.15	95.64 ± 4.02
	MAPE(%)	33.78 ± 0.31	**30.45 ± 1.73**	114.12 ± 0.00	35.59 ± 0.64	41.81 ± 0.46	33.33 ± 0.42	36.21 ± 1.79
HAM	MAE	46.44 ± 0.12	44.26 ± 0.07	97.38 ± 0.48	46.26 ± 0.01	47.54 ± 0.10	**44.16 ± 0.01**	45.04 ± 0.01
	RMSE	77.82 ± 0.73	74.06 ± 0.10	150.66 ± 2.47	77.55 ± 0.01	79.25 ± 0.22	**74.06 ± 0.10**	79.08 ± 1.23
	MAPE(%)	45.57 ± 0.70	44.25 ± 2.04	111.18 ± 3.96	49.44 ± 0.06	50.07 ± 0.06	**43.68 ± 0.20**	44.06 ± 0.07
INN	MAE	71.83 ± 0.48	67.78 ± 0.30	337.43 ± 6.70	86.47 ± 0.02	73.73 ± 0.55	**66.95 ± 0.28**	OOM
	RMSE	104.72 ± 0.59	98.77 ± 1.04	452.25 ± 0.27	133.76 ± 0.11	107.19 ± 0.71	**97.41 ± 0.45**	OOM
	MAPE(%)	29.79 ± 1.13	34.04 ± 3.08	294.18 ± 24.63	37.85 ± 0.91	32.08 ± 0.02	**28.82 ± 1.09**	OOM
KS	MAE	77.06 ± 0.53	83.40 ± 5.34	229.47 ± 0.00	72.68 ± 1.41	85.40 ± 0.05	**70.79 ± 0.35**	190.66 ± 2.64
	RMSE	216.34 ± 0.41	218.00 ± 4.66	332.48 ± 0.00	177.06 ± 2.66	187.31 ± 0.67	**174.04 ± 0.28**	277.37 ± 5.51
	MAPE(%)	97.13 ± 1.41	111.00 ± 17.20	410.19 ± 0.00	99.24 ± 1.87	116.48 ± 0.61	**93.42 ± 0.13**	308.03 ± 0.63
MAN	MAE	106.44 ± 1.13	94.31 ± 0.83	335.71 ± 0.00	103.64 ± 10.23	110.65 ± 0.92	**93.95 ± 1.00**	94.68 ± 2.13
	RMSE	180.81 ± 1.54	169.06 ± 0.26	448.79 ± 0.00	172.31 ± 13.89	184.62 ± 2.27	**163.00 ± 3.21**	168.90 ± 4.47
	MAPE(%)	43.13 ± 1.40	43.34 ± 2.45	279.02 ± 0.00	47.57 ± 4.43	51.47 ± 0.57	**37.68 ± 0.64**	42.02 ± 1.16
MEL	MAE	48.56 ± 0.00	**43.98 ± 1.73**	OOM	58.89 ± 0.02	50.66 ± 0.49	49.02 ± 0.20	46.12 ± 1.29
	RMSE	75.34 ± 0.01	**66.14 ± 3.55**	OOM	89.81 ± 0.46	76.23 ± 0.70	74.99 ± 0.31	72.62 ± 2.27
	MAPE(%)	44.66 ± 0.05	40.38 ± 0.06	OOM	54.22 ± 0.50	53.21 ± 0.56	**33.97 ± 0.06**	39.97 ± 1.00
RTM	MAE	**51.36 ± 0.14**	52.84 ± 1.05	179.34 ± 0.00	64.94 ± 0.05	65.13 ± 0.38	63.44 ± 0.77	55.79 ± 0.08
	RMSE	92.19 ± 0.07	**91.97 ± 0.65**	240.67 ± 0.00	110.95 ± 0.04	106.66 ± 0.68	106.73 ± 1.16	97.29 ± 0.25
	MAPE(%)	**39.55 ± 0.86**	48.88 ± 3.29	349.98 ± 0.00	50.56 ± 0.21	63.69 ± 0.70	49.48 ± 0.75	40.87 ± 1.70
SDR	MAE	101.26 ± 0.42	98.18 ± 2.76	256.61 ± 0.00	93.27 ± 0.06	119.78 ± 2.36	**86.47 ± 0.70**	91.34 ± 2.73
	RMSE	252.60 ± 0.01	249.70 ± 1.91	433.47 ± 0.00	**214.37 ± 0.03**	239.59 ± 2.95	220.15 ± 1.88	235.42 ± 0.34
	MAPE(%)	58.36 ± 1.09	52.17 ± 5.47	255.29 ± 0.00	51.78 ± 0.42	89.07 ± 2.29	49.86 ± 5.91	**41.60 ± 0.36**
SP	MAE	49.19 ± 0.05	**47.78 ± 0.02**	122.39 ± 4.19	52.40 ± 0.01	53.30 ± 0.09	48.28 ± 0.09	48.06 ± 0.17
	RMSE	70.83 ± 0.02	**68.35 ± 0.03**	175.09 ± 12.25	75.17 ± 0.04	75.91 ± 0.14	69.62 ± 0.23	69.28 ± 0.15
	MAPE(%)	39.89 ± 0.29	39.42 ± 0.10	102.80 ± 22.32	44.96 ± 0.02	44.63 ± 0.37	37.82 ± 0.32	**37.33 ± 0.28**
SXB	MAE	78.07 ± 0.07	**75.86 ± 0.22**	261.26 ± 0.00	83.83 ± 0.05	83.48 ± 0.26	76.43 ± 0.07	76.10 ± 0.20
	RMSE	137.75 ± 0.13	**134.97 ± 1.08**	361.96 ± 0.00	146.93 ± 0.03	147.22 ± 0.28	135.84 ± 0.06	136.00 ± 0.61
	MAPE(%)	39.38 ± 0.52	36.88 ± 0.64	223.02 ± 0.00	46.07 ± 0.29	44.95 ± 0.49	38.86 ± 0.28	**36.87 ± 0.37**
STR	MAE	58.31 ± 0.07	55.90 ± 0.33	65.43 ± 3.82	63.52 ± 0.53	67.70 ± 0.29	**55.29 ± 0.12**	OOM
	RMSE	75.53 ± 0.02	72.34 ± 0.67	86.65 ± 5.08	82.43 ± 0.47	86.71 ± 0.16	**71.79 ± 0.33**	OOM
	MAPE(%)	20.16 ± 0.01	**18.61 ± 0.29**	21.85 ± 1.88	22.95 ± 0.11	24.03 ± 0.61	18.78 ± 0.04	OOM
TPE	MAE	134.51 ± 0.20	130.49 ± 1.84	509.35 ± 13.02	138.44 ± 0.00	144.56 ± 0.25	**126.48 ± 0.29**	129.85 ± 0.99
	RMSE	604.37 ± 1.42	606.52 ± 6.88	1002.98 ± 12.83	592.89 ± 0.12	616.55 ± 3.87	**567.78 ± 0.27**	589.03 ± 1.28
	MAPE(%)	45.86 ± 0.49	43.77 ± 0.72	285.05 ± 4.83	44.76 ± 0.06	50.48 ± 0.57	**40.21 ± 1.83**	41.77 ± 1.29
TO	MAE	87.26 ± 0.28	**77.97 ± 0.72**	315.29 ± 0.01	80.61 ± 0.02	83.05 ± 0.09	95.23 ± 1.17	97.61 ± 0.70
	RMSE	149.60 ± 0.47	**128.32 ± 1.66**	421.69 ± 3.47	134.94 ± 0.10	133.55 ± 0.26	150.90 ± 1.60	154.19 ± 0.30
	MAPE(%)	55.62 ± 0.22	**43.26 ± 0.12**	390.80 ± 15.64	46.08 ± 0.69	53.08 ± 1.63	56.75 ± 2.22	63.98 ± 1.64
YTO	MAE	52.35 ± 0.07	**51.04 ± 0.30**	148.49 ± 17.87	86.46 ± 0.04	60.73 ± 0.19	57.26 ± 0.38	51.26 ± 0.14
	RMSE	88.15 ± 0.01	**83.84 ± 0.62**	219.71 ± 22.22	137.24 ± 0.27	95.65 ± 0.01	95.71 ± 0.74	86.95 ± 0.04
	MAPE(%)	38.58 ± 0.02	37.69 ± 3.84	110.54 ± 44.26	65.08 ± 1.92	53.94 ± 2.03	37.79 ± 0.83	**37.62 ± 1.48**
TLS	MAE	257.69 ± 0.41	**255.12 ± 0.04**	264.27 ± 6.31	263.95 ± 0.00	294.82 ± 0.19	255.35 ± 0.01	259.64 ± 0.41
	RMSE	350.19 ± 1.23	342.33 ± 0.54	352.67 ± 9.87	348.26 ± 0.09	408.19 ± 0.21	**340.68 ± 0.33**	351.37 ± 2.05
	MAPE(%)	761.22 ± 0.81	745.90 ± 7.41	792.14 ± 78.55	869.43 ± 1.90	833.62 ± 1.57	749.50 ± 6.72	**730.89 ± 9.58**
UTC	MAE	OOM	66.56 ± 26.30	OOM	48.87 ± 0.11	68.00 ± 0.85	74.93 ± 0.19	**40.09 ± 0.10**
	RMSE	OOM	95.10 ± 17.24	OOM	83.02 ± 0.25	99.18 ± 0.36	122.75 ± 1.30	**72.37 ± 0.03**

	MAPE(%)	OOM	104.15 ± 72.84	OOM	51.11 ± 0.40	90.66 ± 5.26	89.98 ± 4.19	**37.86 ± 3.07**
VNO	MAE	87.69 ± 0.14	83.21 ± 0.40	OOM	74.00 ± 1.45	86.63 ± 0.41	**72.98 ± 0.37**	96.27 ± 0.67
	RMSE	118.99 ± 0.33	113.23 ± 0.07	OOM	100.58 ± 1.74	118.15 ± 0.43	**99.91 ± 0.11**	127.62 ± 0.73
	MAPE(%)	53.52 ± 0.14	48.97 ± 1.46	OOM	41.17 ± 0.65	46.50 ± 0.59	**38.67 ± 1.02**	64.60 ± 1.76
WOB	MAE	53.64 ± 0.29	54.06 ± 3.31	58.06 ± 0.03	61.11 ± 0.00	56.56 ± 0.08	53.34 ± 0.36	**52.71 ± 0.12**
	RMSE	84.29 ± 0.58	83.78 ± 6.06	91.34 ± 0.32	95.94 ± 0.02	85.20 ± 0.03	82.52 ± 0.55	**81.89 ± 0.16**
	MAPE(%)	40.44 ± 0.19	**39.46 ± 1.23**	45.09 ± 0.29	49.95 ± 0.02	48.87 ± 0.60	41.26 ± 0.69	40.60 ± 1.35
ZRH	MAE	OOM	54.55 ± 0.39	OOM	59.12 ± 0.01	58.25 ± 0.02	62.40 ± 10.21	**53.08 ± 0.23**
	RMSE	OOM	77.09 ± 0.55	OOM	83.71 ± 0.00	81.89 ± 0.11	89.30 ± 15.55	**75.22 ± 0.44**
	MAPE(%)	OOM	35.78 ± 2.65	OOM	43.08 ± 0.12	42.26 ± 0.44	46.00 ± 11.94	**35.18 ± 0.34**

Table 9. The performance comparison for seven baseline methods over 31 real-world city datasets with **horizon=12**. The best results in each row are in bold. All experiments are repeated five times, and the mean and standard deviation are reported.

City	Metric	AGCRN	Crossformer	DCRNN	DLinear	FEDformer	GWNet	MTGNN
AGB	MAE	55.05 ± 0.93	**49.30 ± 0.79**	OOM	56.58 ± 0.01	59.64 ± 0.20	55.81 ± 1.34	53.30 ± 0.80
	RMSE	118.56 ± 0.58	**110.08 ± 0.06**	OOM	122.58 ± 0.10	132.58 ± 0.19	117.78 ± 3.74	115.28 ± 0.52
	MAPE(%)	51.73 ± 1.84	**41.13 ± 2.19**	OOM	44.54 ± 0.12	53.19 ± 0.31	46.17 ± 2.80	44.31 ± 3.53
BSL	MAE	74.87 ± 0.54	81.11 ± 3.35	120.92 ± 0.42	76.52 ± 0.22	**66.00 ± 0.70**	97.61 ± 4.34	89.52 ± 0.36
	RMSE	110.37 ± 0.82	122.17 ± 4.15	160.46 ± 2.57	126.12 ± 0.48	**96.52 ± 0.35**	139.03 ± 8.66	126.94 ± 0.56
	MAPE(%)	65.54 ± 0.65	89.04 ± 5.12	183.70 ± 15.83	**62.84 ± 0.30**	69.59 ± 3.62	129.70 ± 6.69	111.43 ± 0.24
BRN	MAE	**53.50 ± 2.10**	54.20 ± 1.85	OOM	56.13 ± 0.01	61.14 ± 0.33	54.13 ± 0.47	84.36 ± 1.26
	RMSE	500.59 ± 21.34	483.08 ± 0.25	OOM	495.21 ± 0.17	489.57 ± 0.68	**479.82 ± 1.60**	553.53 ± 12.87
	MAPE(%)	245.52 ± 4.25	**202.93 ± 62.89**	OOM	290.64 ± 1.52	307.33 ± 16.91	340.44 ± 23.69	513.55 ± 31.06
BHX	MAE	146.03 ± 0.35	102.03 ± 15.50	282.12 ± 13.73	131.89 ± 9.65	140.17 ± 0.70	129.19 ± 2.55	**101.51 ± 0.10**
	RMSE	208.28 ± 0.65	155.04 ± 22.46	384.96 ± 40.10	193.75 ± 6.56	214.60 ± 0.16	192.02 ± 0.76	**154.51 ± 0.86**
	MAPE(%)	109.60 ± 0.28	63.90 ± 4.44	279.89 ± 61.60	90.05 ± 7.98	86.17 ± 0.27	80.07 ± 9.65	**58.53 ± 2.38**
BOL	MAE	**33.38 ± 0.12**	35.32 ± 0.68	45.53 ± 0.92	42.42 ± 0.09	44.03 ± 0.11	39.58 ± 2.22	34.94 ± 0.49
	RMSE	93.57 ± 0.67	93.44 ± 2.09	106.98 ± 0.01	102.71 ± 0.26	105.61 ± 0.31	97.56 ± 0.15	**92.74 ± 0.59**
	MAPE(%)	23.09 ± 0.63	**22.13 ± 0.46**	33.86 ± 1.51	38.07 ± 0.13	33.57 ± 0.61	27.38 ± 1.10	22.27 ± 0.29
BOD	MAE	84.17 ± 0.76	**78.48 ± 2.18**	234.94 ± 3.51	78.76 ± 0.01	87.29 ± 0.40	91.20 ± 1.25	109.35 ± 1.40
	RMSE	138.13 ± 1.65	125.87 ± 0.23	312.43 ± 6.74	**122.38 ± 0.08**	131.87 ± 0.90	139.59 ± 1.34	173.27 ± 0.92
	MAPE(%)	44.29 ± 0.82	**41.19 ± 1.49**	278.32 ± 1.47	54.23 ± 0.18	59.39 ± 0.18	73.00 ± 2.49	74.25 ± 5.28
BRE	MAE	**57.33 ± 0.09**	59.38 ± 0.08	OOM	68.48 ± 0.01	64.98 ± 0.09	58.62 ± 0.06	58.05 ± 0.39
	RMSE	**95.67 ± 0.22**	98.02 ± 0.06	OOM	111.31 ± 0.02	103.81 ± 0.10	96.37 ± 0.06	96.10 ± 0.84
	MAPE(%)	36.89 ± 0.01	**35.19 ± 0.34**	OOM	45.77 ± 0.22	44.95 ± 0.13	38.23 ± 0.29	36.70 ± 0.29
KN	MAE	OOM	52.98 ± 1.36	116.61 ± 0.23	**42.78 ± 0.05**	44.75 ± 0.24	49.44 ± 0.46	57.22 ± 1.96
	RMSE	OOM	83.22 ± 0.84	149.74 ± 0.66	67.32 ± 0.14	**65.83 ± 0.35**	79.03 ± 0.50	86.98 ± 2.16
	MAPE(%)	OOM	**54.05 ± 3.83**	297.19 ± 1.49	70.57 ± 0.27	77.29 ± 3.06	61.92 ± 1.39	89.88 ± 6.97
DA	MAE	56.88 ± 0.01	**56.19 ± 0.15**	OOM	59.55 ± 0.01	61.34 ± 0.24	57.52 ± 0.40	57.85 ± 0.09
	RMSE	88.16 ± 0.02	**86.45 ± 0.70**	OOM	90.81 ± 0.06	90.98 ± 0.34	87.83 ± 1.01	90.13 ± 0.16
	MAPE(%)	53.29 ± 0.09	54.39 ± 6.00	OOM	57.20 ± 0.02	66.56 ± 0.19	55.03 ± 0.35	**52.07 ± 0.04**
ESS	MAE	44.08 ± 0.27	44.91 ± 0.32	179.03 ± 1.10	62.74 ± 0.04	49.97 ± 0.14	41.68 ± 0.05	**41.42 ± 0.27**
	RMSE	65.43 ± 0.68	67.98 ± 0.93	232.52 ± 0.26	92.85 ± 0.16	73.19 ± 0.40	**62.40 ± 0.24**	63.04 ± 0.50
	MAPE(%)	**35.95 ± 1.15**	49.74 ± 5.81	308.94 ± 8.39	55.33 ± 1.11	48.59 ± 1.21	37.57 ± 0.39	38.41 ± 2.97
FRA	MAE	205.72 ± 0.83	205.31 ± 22.15	168.51 ± 1.59	**109.03 ± 36.74**	132.32 ± 1.38	250.59 ± 11.89	386.51 ± 26.96

	RMSE	238.54 ± 0.79	234.38 ± 24.27	209.58 ± 0.03	**132.51 ± 38.23**	165.66 ± 0.15	282.51 ± 11.51	408.51 ± 27.19
	MAPE(%)	73.26 ± 0.25	70.54 ± 5.11	55.50 ± 0.97	**35.27 ± 11.29**	41.22 ± 0.39	85.66 ± 5.06	132.85 ± 10.28
GRZ	MAE	63.46 ± 0.17	**57.03 ± 0.76**	191.54 ± 0.00	71.93 ± 0.01	63.14 ± 0.03	62.37 ± 0.78	59.51 ± 0.28
	RMSE	95.67 ± 0.07	**83.99 ± 0.61**	241.77 ± 0.00	103.75 ± 0.01	91.38 ± 0.00	92.58 ± 0.82	95.09 ± 0.61
	MAPE(%)	109.30 ± 3.51	77.93 ± 5.43	460.74 ± 0.00	81.48 ± 0.13	78.44 ± 0.27	**70.16 ± 4.58**	77.40 ± 2.29
GRQ	MAE	77.05 ± 0.27	**72.08 ± 3.04**	150.91 ± 0.00	72.34 ± 1.76	83.37 ± 2.14	72.33 ± 0.77	82.93 ± 6.04
	RMSE	104.16 ± 0.05	**98.38 ± 2.63**	205.49 ± 0.00	100.03 ± 0.44	115.72 ± 2.69	100.25 ± 0.83	110.28 ± 7.48
	MAPE(%)	41.63 ± 0.69	**35.60 ± 1.85**	117.24 ± 0.00	44.18 ± 2.38	46.94 ± 1.33	38.69 ± 0.88	45.80 ± 2.96
HAM	MAE	47.08 ± 0.23	45.88 ± 0.41	97.51 ± 0.59	48.97 ± 0.02	49.77 ± 0.01	**45.35 ± 0.05**	46.06 ± 0.01
	RMSE	80.67 ± 0.53	78.49 ± 1.08	150.82 ± 2.57	84.20 ± 0.08	83.90 ± 0.06	**77.70 ± 0.01**	82.77 ± 2.08
	MAPE(%)	45.78 ± 0.83	**44.31 ± 3.07**	111.41 ± 4.58	52.32 ± 0.20	53.06 ± 0.21	44.81 ± 0.31	45.29 ± 0.06
INN	MAE	75.51 ± 0.80	72.05 ± 0.00	347.11 ± 2.98	106.94 ± 0.00	80.50 ± 1.00	**69.08 ± 0.43**	OOM
	RMSE	110.87 ± 0.65	106.99 ± 0.06	464.48 ± 8.39	173.49 ± 0.15	117.77 ± 1.61	**100.59 ± 0.79**	OOM
	MAPE(%)	31.27 ± 1.22	36.38 ± 0.72	300.48 ± 37.49	46.75 ± 0.75	34.23 ± 0.15	**27.74 ± 0.06**	OOM
KS	MAE	97.97 ± 0.72	103.19 ± 5.03	227.96 ± 0.00	**78.86 ± 14.60**	104.59 ± 1.19	80.04 ± 0.72	224.87 ± 1.62
	RMSE	244.74 ± 0.48	244.15 ± 6.33	332.77 ± 0.00	**188.64 ± 2.13**	209.20 ± 1.15	195.69 ± 0.26	314.80 ± 4.13
	MAPE(%)	143.62 ± 1.84	156.53 ± 13.18	433.64 ± 0.00	110.39 ± 41.23	162.29 ± 2.84	**109.72 ± 2.55**	407.04 ± 1.53
MAN	MAE	117.20 ± 2.55	109.56 ± 2.55	336.84 ± 0.00	**103.86 ± 9.84**	123.20 ± 0.27	107.88 ± 1.14	107.97 ± 2.60
	RMSE	196.04 ± 4.11	190.75 ± 4.72	450.82 ± 0.00	**173.23 ± 12.91**	198.08 ± 0.51	181.98 ± 4.00	186.65 ± 3.39
	MAPE(%)	47.38 ± 0.86	45.64 ± 2.89	278.97 ± 0.00	46.64 ± 3.55	60.39 ± 0.14	**44.84 ± 1.22**	47.02 ± 1.75
MEL	MAE	64.79 ± 0.03	**54.63 ± 4.48**	OOM	87.95 ± 1.60	67.32 ± 0.31	69.54 ± 1.08	56.73 ± 0.94
	RMSE	101.09 ± 0.06	**81.23 ± 6.52**	OOM	127.60 ± 1.41	97.39 ± 0.04	105.83 ± 1.53	87.59 ± 0.91
	MAPE(%)	55.34 ± 0.02	49.10 ± 5.14	OOM	103.87 ± 4.79	70.74 ± 1.14	**47.17 ± 0.20**	50.21 ± 0.12
RTM	MAE	**57.11 ± 0.15**	58.08 ± 1.24	189.90 ± 0.00	85.30 ± 0.50	82.23 ± 0.36	85.20 ± 1.78	65.01 ± 0.31
	RMSE	99.90 ± 0.34	**98.98 ± 0.53**	250.70 ± 0.00	141.23 ± 0.46	128.17 ± 0.09	136.93 ± 2.26	110.64 ± 0.98
	MAPE(%)	**43.47 ± 0.63**	55.83 ± 11.21	373.03 ± 0.00	65.00 ± 0.87	76.28 ± 0.17	62.30 ± 2.15	46.88 ± 1.28
SDR	MAE	121.60 ± 0.24	129.77 ± 6.53	272.16 ± 0.00	120.00 ± 0.03	152.27 ± 0.77	**104.78 ± 0.34**	119.57 ± 0.47
	RMSE	278.70 ± 0.41	279.99 ± 1.64	440.08 ± 0.00	262.91 ± 0.01	284.60 ± 0.05	**249.10 ± 0.76**	274.87 ± 3.75
	MAPE(%)	67.07 ± 2.54	92.03 ± 14.14	301.72 ± 0.00	68.81 ± 0.28	119.83 ± 1.98	58.89 ± 5.49	**55.51 ± 0.36**
SP	MAE	49.26 ± 0.11	**48.39 ± 0.19**	122.84 ± 4.61	55.70 ± 0.01	54.59 ± 0.03	49.12 ± 0.17	48.70 ± 0.25
	RMSE	70.97 ± 0.13	**69.48 ± 0.62**	175.83 ± 13.31	79.94 ± 0.01	77.28 ± 0.03	70.95 ± 0.34	70.28 ± 0.31
	MAPE(%)	39.90 ± 0.46	38.25 ± 1.95	102.22 ± 24.13	47.80 ± 0.10	46.68 ± 0.32	38.51 ± 0.41	**37.52 ± 0.37**
SXB	MAE	80.33 ± 0.24	**78.32 ± 0.23**	262.40 ± 0.00	94.09 ± 0.05	89.88 ± 0.17	80.19 ± 0.36	78.71 ± 0.29
	RMSE	142.42 ± 0.73	**139.91 ± 0.95**	363.49 ± 0.00	162.33 ± 0.05	156.75 ± 0.32	141.91 ± 0.63	141.32 ± 0.36
	MAPE(%)	40.29 ± 0.54	38.22 ± 0.97	223.27 ± 0.00	51.18 ± 0.37	49.09 ± 0.17	40.96 ± 0.76	**37.89 ± 0.11**
STR	MAE	61.10 ± 0.38	57.78 ± 0.84	71.86 ± 5.01	73.59 ± 0.94	68.30 ± 0.33	**56.40 ± 0.08**	OOM
	RMSE	81.05 ± 0.55	74.94 ± 1.18	96.17 ± 7.97	96.18 ± 1.48	90.50 ± 0.70	**73.46 ± 0.06**	OOM
	MAPE(%)	22.32 ± 0.19	21.28 ± 1.87	25.56 ± 2.23	30.23 ± 0.66	25.02 ± 1.17	**20.36 ± 0.09**	OOM
TPE	MAE	147.54 ± 0.61	**140.98 ± 2.89**	499.95 ± 8.83	163.96 ± 0.02	168.31 ± 0.17	142.71 ± 0.55	142.86 ± 1.83
	RMSE	670.14 ± 0.45	671.29 ± 2.58	996.04 ± 7.32	726.80 ± 0.40	726.73 ± 1.06	**666.42 ± 0.36**	678.02 ± 3.41
	MAPE(%)	54.90 ± 0.81	47.50 ± 4.82	280.63 ± 13.88	52.22 ± 0.17	60.40 ± 0.77	45.94 ± 2.11	**44.89 ± 0.61**
TO	MAE	104.21 ± 0.69	**92.97 ± 2.72**	319.07 ± 5.97	106.30 ± 0.02	105.57 ± 0.66	139.93 ± 0.61	133.21 ± 1.02
	RMSE	178.09 ± 1.38	**154.59 ± 2.71**	424.02 ± 8.74	182.49 ± 0.32	167.92 ± 0.92	231.65 ± 2.32	221.49 ± 0.33
	MAPE(%)	67.26 ± 0.52	**51.82 ± 1.26**	400.37 ± 6.15	59.46 ± 1.38	69.03 ± 0.16	82.81 ± 0.41	87.58 ± 0.12
YTO	MAE	57.16 ± 0.24	**56.66 ± 1.67**	173.86 ± 9.18	125.57 ± 0.08	74.96 ± 1.04	70.85 ± 0.31	60.80 ± 0.21
	RMSE	94.09 ± 0.11	**90.75 ± 1.41**	253.42 ± 9.38	191.73 ± 0.25	114.34 ± 1.97	115.62 ± 0.60	100.24 ± 0.45

	MAPE(%)	44.91 ± 1.53	**43.39 ± 4.12**	128.97 ± 45.99	110.73 ± 1.97	73.18 ± 1.56	44.39 ± 0.43	43.66 ± 0.17
TLS	MAE	257.57 ± 0.47	255.17 ± 0.18	264.44 ± 6.13	264.06 ± 0.00	298.73 ± 0.16	**255.11 ± 0.01**	257.76 ± 0.30
	RMSE	348.95 ± 1.83	342.20 ± 1.02	352.99 ± 9.76	348.31 ± 0.02	408.17 ± 0.23	**340.28 ± 0.14**	346.83 ± 0.97
	MAPE(%)	746.15 ± 8.05	758.84 ± 36.31	791.75 ± 78.81	867.12 ± 0.77	842.35 ± 8.43	**745.07 ± 5.77**	746.42 ± 0.82
UTC	MAE	OOM	55.13 ± 4.47	OOM	57.76 ± 0.72	73.45 ± 1.74	73.81 ± 0.57	**40.36 ± 0.11**
	RMSE	OOM	89.55 ± 1.67	OOM	97.52 ± 1.24	109.98 ± 1.82	123.95 ± 2.23	**75.86 ± 0.31**
	MAPE(%)	OOM	80.70 ± 18.35	OOM	64.93 ± 0.77	105.19 ± 2.55	93.66 ± 5.16	**40.85 ± 2.82**
VNO	MAE	96.16 ± 0.15	91.31 ± 0.19	OOM	81.71 ± 4.58	98.47 ± 0.10	**80.01 ± 0.59**	105.63 ± 1.55
	RMSE	130.79 ± 0.34	124.11 ± 0.44	OOM	111.45 ± 2.76	131.28 ± 0.02	**109.93 ± 0.18**	141.02 ± 1.91
	MAPE(%)	63.46 ± 0.21	58.74 ± 1.31	OOM	47.91 ± 8.96	60.49 ± 0.08	**46.17 ± 1.50**	77.00 ± 2.31
WOB	MAE	56.73 ± 0.26	**53.73 ± 0.93**	65.57 ± 0.76	69.50 ± 0.05	61.35 ± 0.11	57.82 ± 0.18	56.30 ± 0.43
	RMSE	90.97 ± 0.48	**83.48 ± 2.09**	104.61 ± 1.31	111.82 ± 0.01	92.81 ± 0.13	91.64 ± 0.05	89.31 ± 0.69
	MAPE(%)	42.48 ± 1.04	**41.16 ± 3.01**	53.82 ± 1.86	55.70 ± 0.59	54.53 ± 0.12	44.75 ± 0.61	43.14 ± 1.57
ZRH	MAE	OOM	56.73 ± 0.03	OOM	65.62 ± 0.00	65.29 ± 0.20	61.77 ± 4.83	**54.71 ± 0.20**
	RMSE	OOM	80.80 ± 0.23	OOM	94.06 ± 0.01	91.54 ± 0.20	88.03 ± 6.88	**77.75 ± 0.44**
	MAPE(%)	OOM	36.22 ± 0.43	OOM	47.73 ± 0.13	48.20 ± 0.21	46.66 ± 8.57	**35.87 ± 0.06**

G. Author Statement

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